

SYSTEM SEGMENT SPECIFICATION

KEYLOCKS / EULYNX

Component: ECMUX
Solution Element: Virtual Interlocking OBB 1.0.0
Solution: Digital Interlocking OBB
Country Austria

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
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APPROVAL		
	Name and Role/Job Title	Esignature
Written by	A. Korlevic / M. Ionita, DEV / DES	
Verified by	R. Al Ali, QAM	
Approved by	E. Glaser / R. Holzmann, PDA / ARC	

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LOG OF CHANGES			
Revision	Date	Author	Change Description
ED01 PD01	2025-12-04	A. Korlevic, DEV M. Ionita, DES	Initial proposal. Contains only Line Keylock server version (KLL server).
ED01 PD02	2026-01-23	A. Korlevic, DEV M. Ionita, DES	Contains fixes for review comments for PD01. Added Line Keylock client version (KLL client).

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1 Scope

1.1 Identification

This System Subsystem Specification defines requirements for the element Keylock of the component ECMUX within the subsystem VIL of the OBB Digital Interlocking.

The requirements of the component ECMUX are represented by a set of SSS documents:

- the ECMUX generic SSS [5] describing the core functions of the SW component like the common element repository, the interface to other interlocking components, communication protocols, diagnosis and maintenance functionality
- several element specific SSSs, which are addressing one specific element/functional feature of ECMUX

This SSS is specific for Keylock and therefore extends the ECMUX generic SSS [5] with Keylock specific behavior. This SSS provides the requirements for the functional element Keylock derived from system (or sub-system) level requirement documents DIL-CU SSS [9] and VIL SSS [7]. Other input documents are Elektra Funktionelle Spezifikation Streckenschlüsselsperre mdref:FUNK-SPEC-KEYLOCK and HW Solution for Keylocks with DCA 3.0 [16].

For the general structure of requirement documents in the scope of OBB Digital Interlocking, which are relevant in ECMUX scope, how ECMUX SSSs are embedded into this structure and how the traceability is planned, see the overview in Chapter 1 "Introduction" of ECMUX generic SSS [5].

The "Keylock SW" identifies the layers and modules of all software within the ECMUX, which are working together to fulfill the functionality for a Keylock controlling module.

The architecture and design of generic ECMUX is part of the System Subsystem Design Description document ECMUX generic [3], the architecture and design of the element Keylock is part of the Software Design Description document ECMUX SW Keylock / EULYNX [4].

1.2 System Overview

1.2.1 ECMUX Overview

ECMUX is a component to control trackside (outdoor) railway equipment via Eulynx standardized interfaces. Following the Eulynx reference architecture, the ECMUX is part of the Electronic Interlocking (EIL) subsystem and there acts as interface component between an interlocking core logic and the Eulynx subsystems for outdoor element control, the Object Controllers (OCs). The main task of ECMUX is to command and control the Object Controllers assigned to it.

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The ECMUX is not a stand-alone product, it is designed to be part of the ELEKTRA Digital Interlocking (DIL) system. Together with the ELEKTRA Core and Diagnosis components it implements the Virtual Interlocking (VIL) subsystem within ELEKTRA Digital Interlocking (DIL) system.

Within DIL, ECMUX acts

- as a protocol translator - ELEKTRA CC interface protocol is translated to Eulynx protocols, and
- as multiplexer - it's able to connect many OCs and offer one single interface connection to the interlocking core logic

The ECMUX is intended to be executed in an virtualized environment within a centralized data center. The operated Object Controllers are located on-site in a station next to the track to control all hardware devices nearby. Therefore, high distances to the OCs are the general case, so all connectivity is IP-based.

1.2.2 System OBB Digital Interlocking

The OBB Digital Interlocking (DIL) defines a new generation of interlocking systems based on the Eulynx reference architecture and follows the basic ideas of:

- a centralized interlocking architecture, with interlocking instances running virtualized on server environment
- standardized, Eulynx compliant element control via on-site elements, the Object Controllers
- using Eulynx standardized interfaces for communication between all components according to Eulynx reference architecture based on RaSTA protocol / IP
- clear separation of safety relevant interlocking functionality from diagnosis and maintenance components
- high level of cybersecurity resilience

DIL operates in a highly controlled, safety-critical railway environment. It is designed to function reliably under varied operational, environmental, and network conditions.

DIL refers to the overall system consisting of the server-based central units (server hardware plus Virtual interlocking - VIL), the IP network, the object controllers and the corresponding outdoor system. VIL - Virtual Interlocking - consists of the main parts Interlocking Core Logic, the ECMUX interface host, the RCU interface host and the Diagnosis component.

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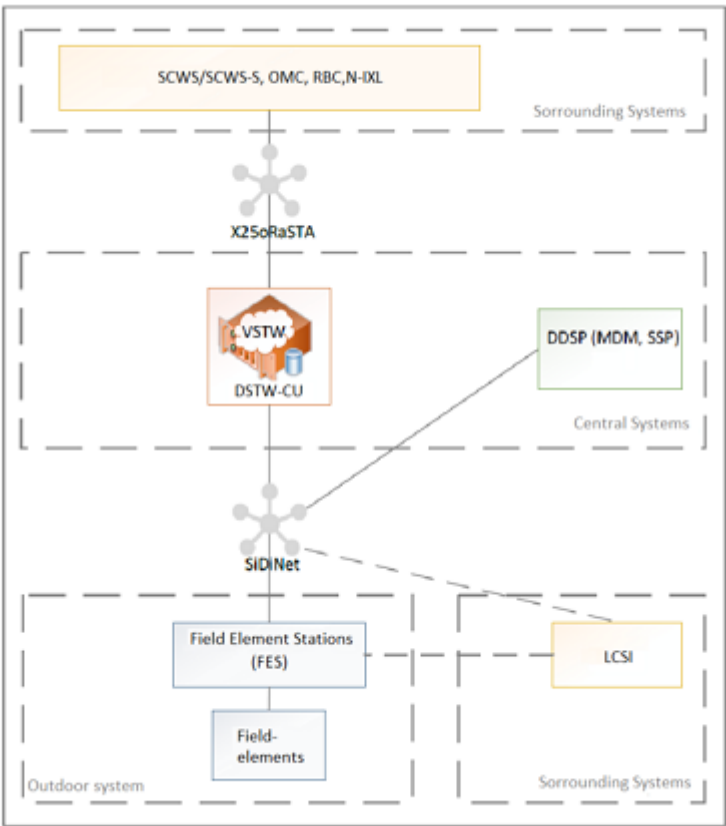


Figure 1: Digital Interlocking System (DIL) Overview

The DIL must exchange data with the existing peripheral systems such as the Signal Controlled Warning System (SCWS/SCWS-S), operations control centre, Radio Block Centre (RBC), neighbouring Interlockings and level crossings, as well as the maintenance and diagnostics system (MDM) and the services of the security services platform (SSP). The MDM and SSP system will be combined in the so-called Diagnosis and Data Service Provider (DDSP) system.

The field element station (FES) is part of the outdoor subsystem and the link between the Virtual Interlocking core logic and the outdoor subsystem elements. It is used to accommodate all decentralized installed electronic components for the security technology (object controller) and to connect to the outdoor field elements.

The OCs are connected to the operating ELEKTRA Virtual Interlocking Subsystem VIL (described in Subsystem SSS Virtual Interlocking [7] and Subsystem SSDD Virtual Interlocking [8]). ECMUX represents the interface host of VIL to the OCs.

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1.2.3 ECMUX Component

The ECMUX component is the ECMUX application software executed in an virtualized environment of TAS Platform. It is required to be a SIL4 application and implemented on security level SL3. The ECMUX implementation is done according to the Eulynx standard reference architecture [13].

As interface component, ECMUX is capable of receiving commands from the interlocking core logic, translates them to Eulynx protocol commands, forward them to the outdoor elements and send back state and diagnostic information. In its role as multiplexer, ECMUX is able to connect up to 400 OCs to the interlocking core.

To connect the indoor ECMUX and outdoor OC components, the standardized SW protocol set of Eulynx is used [13]. The OCs represent the Eulynx subsystems of type outdoor elements.

Because the Eulynx reference architecture defines a separation of command and control interface (SCI, Standard Communication Interface, implemented in SIL4 context) from diagnosis and maintenance (SDI, Standard Diagnostic Interface, and SMI, Standard Maintenance Interface) and from SSI (Standard Security Interface), just the SCI protocol is of interest for the ECMUX.

ECMUX is intended to be executed in an virtualized environment within a data center with possibly high distance to the OCs / field elements. All connectivity to other components is therefore IP based.

The ECMUX component implements the interfaces for all types of Eulynx subsystems (object controller) for control of peripheral elements. An object controller is of a specific type according to the Eulynx subsystems:

- OC-P - Point subsystem
- OC-LS - Light signaling subsystem
- OC-TDS - Train detection subsystem
- OC-LX - LevelCrossing External subsystem
- OC-IO - Generic IO subsystem

An interlocking system can have one or more ECMUXes (up to 4 according to system limits) controlling up to 400 OCs each:

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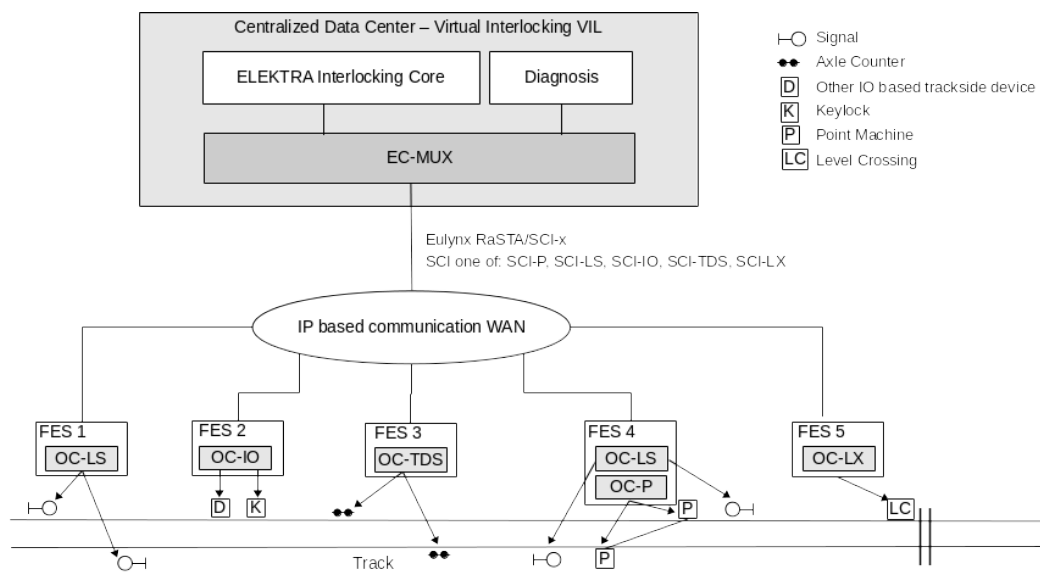


Figure 2: ECMUX Overview

- ECMUX is the element multiplexer. It is an indoor component located in the control center and able to handle many processing units (OCs) along the track. It has the task to provide an interface for receiving commands and sending state and failure information back to the superior interlocking logic.
- OCs are the object controllers inside the FES (field element station, the cabinet). It is located next to the track operating the elements nearby. The OCs represent Eulynx outdoor subsystems and are connected to the ECMUX via Eulynx SCI protocols.

1.2.4 ECMUX generic software libraries

The software of an ECMUX component always consists of some generic parts (ECMUX SW GEN libraries), which are common for all country specific implementations, and a country specific part extending the generic base libraries.

ECMUX SW GEN is the generic software library, which defines a basic system to operate OCs by implementing generic versions of the indoor component ECMUX and implementing interfaces to the OCs.

This basic component provides all country independent functionality for ECMUX SW specific solutions:

- common functionality
- base communication
- diagnosis system
- startup and shutdown mechanisms
- build mechanisms
- virtual machine support
- safety concept

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ECMUX SW GEN is deployed as some Java software libraries, which are supposed to be included in country specific ECMUX SW implementations. It is assumed, that the generic libraries are always embedded into a country specific solution adding specific functionality.

ECMUX SW GEN implements the following generic software modules and supports the following interfaces:

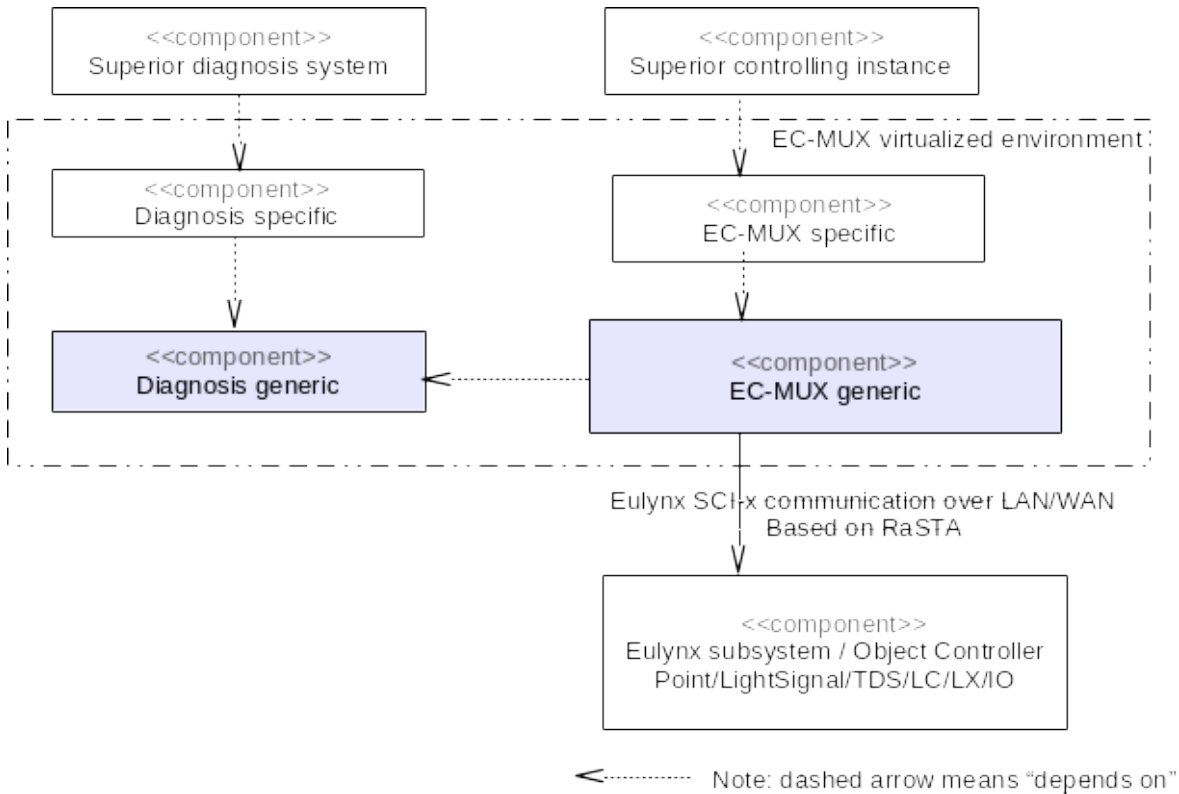


Figure 3: ECMUX SW GEN: main software modules and interfaces

- ECMUX command interface: the ECMUX provides a country independent software interface to allow any controller using the ECMUX SW system to operate trackside elements and receive state and failure information
- Interface Eulynx SCI-x - IP based communication over WAN/LAN / single-message-safe principle using RaSTA: the protocol SCI-x is the Eulynx application layer interface for communication between ECMUX and OCs. On transport layer, according to Eulynx it is sent via the RaSTA [15] protocol
- Module ECMUX generic: software channels for the element controller multiplexer - operates multiple OCs, provides generic functionality to send/receive command/state information to/from the correct OCs
- Module Diagnosis generic: common instance to collect all generated diagnosis data from the whole ECMUX SW system

The components supposed to extend ECMUX SW GEN in any country specific implementation are:

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- ECMUX specific: implementation of the ECMUX command interface, translation of incoming commands from the superior controller to ECMUX commands, vice-versa for element states and failures sent back
- Diagnosis specific modules: implementation of specific translation of messages to/from a superior diagnosis system

The element specific diagnosis strategy and design are in detail described in the Software Design Description document ECMUX SW Keylock [4].

There are two types of diagnosis:

- structured diagnosis mechanisms as defined in System Subsystem Design Description document ECMUX SW GEN [3]
 - Element Aspect Diagnosis
 - Rejected Command Diagnosis
- plain text diagnosis specific for element behavior described in Software Design Description document ECMUX SW Keylock [4] and in sub-chapter *Requirement Language*

1.2.5 Keylocks Overview

A keylock is a lockable control element in the safety technology of the railway. Unlike the normal lock, it isn't possible to simply remove the key because it is mechanically blocked. The key can only be removed from the housing if it is released by the interlocking control.

When removed, the key can be inserted into the control element (usually a point) to unlock it and switch its position. While the control element is in the new position, the key is locked within the control element. Only when the control element (the point) is returned to the normal position, can the key be removed and put back into the keylock housing where it will be immediately locked.

While the key is removed from the keylock housing, the interlocking control prevents the point from being used by a second vehicle. The key thus becomes a token with which either the point is locked and used by the interlocking control, or to which the interlocking transmits the control (e.g. to make a connecting run out of the track via the point).

Depending on the location, two types of keylocks exist:

- Station keylock (KLS) is located in the own station
- Line keylock (KLL) is located on the line between two stations

HW of a Line keylock is connected to both neighboring stations but in different configurations:

- server version: One of the neighboring stations (the operating station) will have a server connection to the KLL HW. The KLL is in this station configured as a Line keylock operated by the own interlocking (German: Schlüsselsperre in der eigenen BÜA bedienbar). This KLL can be commanded from the GUI of the interlocking.
- client version: The second neighboring station will have a client connection to the KLL HW. The KLL is in this station configured as a Line keylock operated by the neighboring interlocking

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(German: Streckenschlüsselsperre die nicht in der eigenen BÜA bedienbar ist aber meldeseitig verwaltet wird). From GUI no line keylock specific commands can be entered by the operator, only the current state of the element can be observed. In ELEKTRA there are 2 configuration possibilities for a line keylock client:

- KLL connected via Relay interface
- KLL connected via X25 interface

In ECMUX only KLL client connected via Relay interface is implemented.

For more details about specific types of keylock see following sub-chapters.

Communication between ECMUX and CC takes place via the CC Interface (see [6]).

1.3 Document Overview

This document is the System Subsystem Specification for the Keylocks SW subsystem of the EC-MUX software library that specifies only the OSI layer 7 (application layer). The underlying OSI layers are specified in [5] and in [3].

Chapter 1 defines the scope of the document, identifies the system and explains ECMUX and the Keylocks subsystems.

Chapter 2 lists all references, terms, abbreviations and requirement language.

Chapter 3 of this document describes the requirements for the Keylock functionality within the ECMUX library.

Chapter 4 List of all requirements from chapter 3 Requirements.

Chapter 5 Index of all requirements from chapter 3 Requirements.

1.4 Intended Audience

The target audience for this document is:

- Product Design Authority (PDA)
- Solution Engineering Manager (SEM)
- Safety Assurance Manager (SAM)
- Verifier (VER)
- Validator (VAL)
- Integration Verification, Validation, Qualification Manager (IVVQM)
- Architect (ARC)
- Requirements Manager (RM)
- Product Line Manager (PLM)
- Quality Assurance Manager (QAM)
- Security Assurance Manager (SecAM)
- Project Security Manager (PSM)

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- RAM Assurance Manager (RAM-AM)

1.5 Evolution of the document

The document is written iteratively according to the Agile Process defined in the SEA Team Charter [1]. This means that several document editions can be released before the functionality is completely described/defined. For the details on what is included in which edition of the document please see the section *History* below.

After the final edition containing all information for the planned functionality is released, it will be updated accordingly in case of requests for changes of the system. For type and reasons of changes please refer to the section *History* below.

1.5.1 History

1.5.1.1 Edition 01 PD01

This document proposal contains requirements that define the features of the Line Keylock Server version. Station Keylock Single Lock and Line Keylock Client version are not included.

1.5.1.2 Edition 01 PD02

This document proposal contains incorporated comments from PD01 and additionally Like Keylock Client version.

1.5.1.3 Edition 01 PD02

This document proposal contains fixes for review comments of PD01. Additionally, Line Keylock Client version is included. Station Keylock Single Lock is not included.

1.5.2 Open issues

- Station Keylock single lock is not included
- Replace SEA with ECMUX in Chapter 2 and Chapter 3
- Add a reference to finally used versions of DCA3.x (and DCA2.x) to *System Overview*
- Traceability of Eulynx specifications is not included in the document
- Security aspects to be considered in later phases of the project
- IDD CC/DGP - ECMUX, SEA SW GEN SSS, SEA SW GEN SSDD to be updated in ELEKTRA 4 context

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1.5.3 Distribution

A new version (proposal) of this document has to be distributed to all functions listed in the section Intended Audience. A new edition of this document has to be archived (online archive and conventional archive). Due to possible large numbers of involved parties the online archival of proposals is recommended. A copy for information can be downloaded from online archive by everyone in the project, who has read privilege.

1.5.4 Configuration Management

This document is generated by the Doc Maven Plugin. It is composed of textual building blocks; the content structure and metadata are defined in a "document definition" XML file. These source files are archived in the Git repository of Hitachi Rail GTS Austria, where all changes are tracked. The generated document is published on the project's website (SEA Project Site) as XHTML and PDF. Formal review proposals and released versions are archived in the Hitachi Rail GTS Austria central document repository in PDF format.

Git URL: <https://bitbucket.de.gts/scm/sea/sea>
Git commit: 081e99499d
Document definition file: /doc/src/doc-definition/sss_keylocks.xml

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2 References and Terms

Remark: Documents are mentioned with edition only if they are input documents. Documents without edition information are for information only.

2.1 SEA Generic Documents

[1]	Team Charter SEA, 3BU 61400 0137 QMAPA, Ed. 04
[2]	HITACHI-GLOSSARY, 3BU 15000 8008 WDAPO, Ed. latest, Hitachi Rail Austria - Glossary

2.2 SEA Design Documents

[3]	SEA SW GEN SSDD - System Subsystem Design Description, 3BU 61400 0200 DSAPA, Ed. 01 (planned)
[4]	SDD Keylocks, 3BU 61400 0186 PEAPA, Ed. 01 (planned)

2.3 SEA Specification Documents

[5]	SEA SW GEN SSS - System Subsystem Specification, 3BU 61400 0200 DTAPA, Ed. 01 (planned)
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2.4 Elektra Documents

[6]	IDD CC/DGP - ECMUX, 3BU 15000 9130 PBAPA, ED. 06 - To be updated
[7]	Subsystem SSS Virtual Interlocking, 3BU 15000 6955 DTAPA, Ed. 01 (planned)
[8]	Subsystem SSDD Virtual Interlocking, 3BU 15000 6955 DSAPA, Ed. 01 (planned)
[9]	SSS Digital Interlocking, 3BU 17300 1255 DTAPA, Ed. 01 (planned)
[10]	Funktionelle Spezifikation, Streckenschlüsselsperre, 3BU 20112 AAAA DSAPC, Ed. 07
[11]	System Requirements Specification ELEKTRA OEGB Betriebliche Funktionen, 3BU 62300 0002 DPAPA, Ed. 05

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2.5 External documents

[12]	[EULYNX Requirements specification for Generic IO subsystem], EULYNX Requirements specification SCI-IO, Eu.Doc.45, Baseline 4.3 (0.A)
[13]	[EULYNX System Definition], EULYNX System Definition, Eu.Doc.7, Baseline 4.2 (0.A)
[14]	[EULYNX Interface specification SCI-IO], EULYNX Interface specification SCI-IO, Eu.Doc.46, Baseline 4.0 (2.A)

2.6 Protocols

[15]	RaSTA, Rail Safe Transport Application, V03.03, DIN VDE V 0831-200, 3BU 01500 0006 PBAPA, Ed. 01
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2.7 Hardware

[16]	EL-SYS-23-505_Schluesselsperren_DCA, EL-SYS-23-505_Schluesselsperren_DCA, Ed. 01
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2.8 Structure of Requirements Definition

This subsection defines the structure of one single requirement. The keywords and their explanations are shown below:

- **Identifier:** Unique identifier of the requirement
- **Type:** A classification of the different requirement types; the following keywords are defined:

```

`CSR` Customer Specific Requirement
`DAT` Data Requirement
`DOR` Document Requirement
`DPR` Dependability Requirement
`ECR` Environmental Conservation Requirement
`FUR` Functional Requirement
`GCR` General Constraint Requirement
`MCR` Management and Commercial Requirement
`MDR` Maintenance and Diagnosis Requirement
`MIR` Man-Machine Interface Requirement
`MPR` Management and Production Requirement
`OAR` Operational Aliveness Requirement
`PER` Physical/Environmental Requirement
`PHR` Prevention of Hazards Requirement
`PRR` Performance and Real-Time Requirement
`PSR` Product Startup Requirement
`RSR` Resource Support and Consumption Requirement
`SAR` Safety Requirement
`SCR` Supplier to Customer Requirement

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`SER`	Security Requirement
`TIR`	Technical Interface Requirement
`TVR`	Testing and Validation Requirement

- **Synopsis:** A brief and significant description of the requirement.
- **Description:** Complete description of the requirement. Includes, when needed, layouts (screen layouts in principle - no screen dumps), formats, and protocols for interfaces.
- **Satisfies (Origin):** Source of the requirement (feature list, reference to other requirements or internal). It is a precise backward reference of this requirement to its origin. Per convention, references to the Feature List SEA are marked FR:SEA:xxx.
- **Status:** The current status of the requirement, with the following keywords defined:
 - in work: new requirement which is not yet described completely
 - draft: requirement is written down and ready to be reviewed, it is not yet approved
 - approved: requirement was reviewed by all affected groups and is confirmed
 - descoped: Requirement delayed to a later release
 - to be descoped: requirement is planned to be descoped, no formal approval for changing the state yet
 - to be deleted: Requirement is planned to be deleted. No formal approval for deleting by CCB yet
 - deleted: Requirement is no longer valid and will not be implemented at all. Has to be agreed by CCB
- **Stability:**
 - stable: will probably not change during the life time of the product
 - volatile: subject to change
 - instable: subject to change frequently
- **Target Release:** Scheduled for which release
- **Priority:** Relative importance of fulfilling this requirement from a functional point of view (used for incremental deliveries). The following keywords are defined:
 - Mandatory: Requirement required by applications. If requirement is not provided, the fact and the consequences have to be documented in the validation report. The application has to decide if the product is accepted or not in absence of this requirement.
 - Optional: Enhancement of the product may be implemented in this or a later release. Absence does not make the product unacceptable.
- **Originator:** "Equipment Practice", "Product Management", "Quality", "RAM", "Safety", "Standards /Legal", "Technics (R&D)"
- **Allocation:** Allocates requirements to the system, subsystem, component, or product environment as described in a product breakdown structure. The appropriate keywords have to be defined by the project.
- **Verification Method:** "Analysis", "Demonstration", "Inspection/Review", "Test"
- **Verification Step:** The following verification steps are defined:

Review
Code Reading
Analysis
Module Test
Sublt Test
Integration Test
System Test

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Performance Test
Qualification Test
Acceptance Test
Field Test
Tool Test
Safety Approval

- **Comment:** Any additional information if necessary. (e.g. reason for deleting a requirement)

2.9 Requirement Language

This subsection defines the language used in requirements and SUBIT test cases. Note that all actions are specific to the element for which the requirement or test case is written, e.g. by received a command it is meant the command received for the element in question, by reacting or not reacting to an input it is meant that SEC is doing an action on behalf of the element in question, without considering any other elements, unless this is explicitly mentioned.

The element state mentioned is the SEA internal element state as defined in State Machine Diagram.

General:

- “SEA shall do nothing”:

SEA shall do nothing in regard to the element in question:

- the state of the element is not changed
- element specific timers are not changed
- no command is sent to the HW interface or to another internal SEA component
- no message is sent on the CC and DGP interface

- “Action within t ”:

Action is performed within the specified time interval from the last action.

- “Action after t ”:

Action is performed specified amount of time after the last action.

- “Action after t from *ReferencePoint*”:

Action is performed specified amount of time after the reference point.

- “Action when reaching state *State*”:

Action is performed when the SEA internal element state has changed to the specified state.

- “SEA shall set the current element state to *State*”:

SEA shall set the SEA internal element state to *State*.

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- “SEA shall switch to *State*”:

SEA shall

- set the SEA internal element state to *State*
- command the output or check-back port(s) corresponding to the element state *State* (as defined in the mapping table)

- “the current element state is *State*”:

SEA internal element state is *State*.

- “the element state has changed to *State*”:

SEA internal element state changed.

- “SEA shall execute ECMUX safety reaction *SafetyReaction*”

SEA shall execute one of the safety reactions defined in SSDD SEA GEN [3], Table “ECMUX Safety Reactions”.

- “an *Element* configured as *Type*”:

The element configuration corresponds as defined for the specified type.

- “initializing the element”

At SEA start-up, one of the performed actions is to read the xml database file containing station data and setup the element repository. Each element contained in the station data shall take on an initial state according to its element type configuration.

ECMUX booting:

- “shutting down ECMUX”

Depending on the environment on which the ECMUX is running on, in the case of a physical environment this step will cause a shut down of the ECMUX software by triggering a reboot of the hardware or a shut down of the ECMUX in the case of a virtualized environment.

- “starting up ECMUX”

Wait until the ECMUX is started again (for physical environment hardware is booted up again, for virtual environment ECMUX is started). This includes also establishing the connection between ECMUX and HW (OEC, OC...) or simulation.

Internal communication between SEA elements:

- “changing the state of *Element*”:

The element (for which the requirement is written) detects a state change of another element.

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DGP Interface:

- “SEA shall report the diagnosis aspect message *Aspect* to DGP”:

Send e_sea_dgp_elem_aspect_reply with the given aspect (corresponding to the element state) from SEA to DGP via DGP signal s_elx_dgp_01b2_diagnose.
- “SEA shall report the *Fault* diagnosis device fault message for the *Device* device to DGP”:

Send e_sea_dgp_elem_hw_fault for the element with the given faulty device (e.g. IOC) from SEA to DGP via DGP signal s_elx_dgp_01b2_diagnose.
- “SEA shall report the *Fault* diagnosis device fault message to DGP”:

Send e_sea_dgp_elem_hw_fault for the element (no device specified) from ECMUX to DGP via DGP signal s_elx_dgp_01b2_diagnose.
- “SEA shall report the *Fault* diagnosis device fault repair message for the *Device* device to DGP”:

Send e_sea_dgp_elem_hw_repair for the element with the given faulty device (e.g. IOC) from SEA to DGP via DGP signal.
- “SEA shall report the *Fault* diagnosis device fault repair message to DGP”:

Send e_sea_dgp_elem_hw_repair for the element (no device specified) from ECMUX to DGP via DGP signal s_elx_dgp_01b2_diagnose.
- “SEA shall report the rejected command diagnosis to DGP”:

Send e_pc_dgp_elem_cmd_info from ECMUX to DGP via DGP signal s_elx_dgp_01b2_diagnose.
- “SEA shall report the plain text diagnosis message *MsgText* to DGP”

Send e_sea_dgp_string containing the message text *MsgText* from ECMUX to DGP via DGP signal s_elx_dgp_01b2_diagnose.
- “When receiving an aspect query from DGP”:

refers to receiving the e_dgp_sea_elem_aspect for the element from the DGP. SEA responds by one or more e_sea_dgp_elem_aspect_reply messages for the addressed element.

CC Interface connection and fault handling:

- “SEA shall send the fault notification sum_fault with fault type e_if_error to CC”

Send the fault information for the element from ECMUX to CC via the s_epa_gcx_C20B_sum_fault /s_epb_gcx_D30B_sum_fault CHILL signal.
- “SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC”

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Send the fault information for the element from ECMUX to CC via the s_epa_gcx_C20C_sum_fault_rep/s_epb_gcx_D30C_sum_fault_rep CHILL signal.

- “establishing the connection to CC”

This includes successful completion of all phases required for an operational connection, like version handshake, update_history and full_scanner phases and general inquiry. See [6] for details. Any commands sent by CC during update_history phase are ignored, for details see Chapter TODO:SEA-6949

- “initiating the connection establishment to CC”

CC connection establishment is done in several phases (like version handshake, update_history and full_scanner phases and general inquiry). This step includes all phases till the beginning of update_history phase. See also “receiving a command from CC during update_history phase”, “answering the general inquiry from CC” and “establishing the connection to CC”. For full operational connection all phases need to be completed.

- “answering the general inquiry request from CC”

CC connection establishment is done in several phases (like version handshake, update_history, full_scanner phases and general inquiry). This step handles answering the *general inquiry request* (3d86_send_elem signal with *QueryType* REQUEST_STATE) received after the general inquiry phase has started (3d86_send_elem signal with *QueryType* GENERALABFRAGE_BEGINN). After the step is finished, full operation is reached. For handling of phases before general inquiry begin see “initiating the connection establishment to CC”.

- “losing the connection to CC”

ECMUX does no longer see the CCx. This may either be caused by network connectivity loss, or by CCx reboot. Implicitly, while there is no connection to the CC, no communication (receiving commands or reporting) will be possible.

Connection loss to CC is detected by at least one of the following:

- CCx lifetick supervision failure
- CCx does not respond to ECMUX reactivity supervision signals
- CC-IF-RASTA connection down
- “CC connection state is
[CONNECTED|NOT_CONNECTED|CONN_INITIALIZATION|UPDATE_HISTORY|FULL_SCANNER]

CC connection state CONNECTED means that the connection between ECMUX and CC is fully operational, the connection procedure as explained under “(re)establishing the connection to CC” has been performed.

CC connection state NOT_CONNECTED means that there is no connection between ECMUX and CC, the CC connection was never available or the disconnection procedure as explained under “losing the connection to CC” has been performed.

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CC connection state CONN_INITIALIZATION means that the connection establishment procedure has been initiated and signal exchange till update_history phase is in progress (like version handshake). This ends with the *update_history begin* (update_sec signal with *isBegin=true*) when the UPDATE_HISTORY is started.

CC connection state UPDATE_HISTORY means that the connection establishment procedure has been initiated and all phases before update_history phase have been successfully finished (like version handshake). *update_history begin* (update_sec signal with *isBegin=true*) signal has been received. This phase ends with the *update_history end* (update_sec signal with *isBegin=false*) when the FULL_SCANNER is started.

CC connection state FULL_SCANNER means that the connection establishment procedure has been initiated and all phases before have been successfully finished (like version handshake, update_history...). *update_history end* (update_sec signal with *isBegin=false*) signal has been received and ECMUX answered with the full_scanner signal. *general inquiry begin* (send_elem signal with *QueryType* GENERALABFRAGE_BEGINN) when the GA_STARTED is started.

CC connection state GA_STARTED means that the connection establishment procedure has been initiated and all phases before general inquiry phase have been successfully finished (like version handshake, update_history and full_scanner). *general inquiry begin* (send_elem signal with *QueryType* GENERALABFRAGE_BEGINN) signal has been received, ECMUX is waiting for *general inquiry request* (send_elem signal with *QueryType* REQUEST_STATE) from CC.

- “the connection to CC has been lost”

ECMUX lost the connection to the CC. See also “losing the connection to CC”.

CC interface

- “receiving a command from CC”

SEA receives a command from CC via the defined restraint-safe s_gca_gpx_2C80_peri_out /s_gcb_gpx_3D80_peri_out CHILL signal or via the defined execution-safe s_gca_gpx_2C81_2chan_out/s_gcb_gpx_3D81_2chan_out CHILL signal. Note that the command can be received also before the connection is fully established, i.e. before the general inquiry is finished. The command can have any command source.

- “receiving the command *Command* from CC”

SEA receives a command from CC via the defined restraint-safe s_gca_gpx_2C80_peri_out /s_gcb_gpx_3D80_peri_out CHILL signal or via the defined execution-safe s_gca_gpx_2C81_2chan_out/s_gcb_gpx_3D81_2chan_out CHILL signal and process it. Note that the command can be received also before the connection is fully established, i.e. before the general inquiry is finished. The command has not been generated due to CC update history mechanism (command source is different than e_boot_update_pc and e_recover_update_pc).

- “receiving the command *Command* with the command source *CmdSource* from CC”

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SEA receives the command from CC via the defined restraint-safe s_gca_gpx_2C80_peri_out /s_gcb_gpx_3D80_peri_out CHILL signal or via the defined execution-safe s_gca_gpx_2C81_2chan_out/s_gcb_gpx_3D81_2chan_out CHILL signal and process it. Command source in the signal is set to the defined *CmdSource*.

- “receiving an unknown command for a known element from CC”

SEA receives a command from CC that is not defined for the given element. Note that the connection is fully established meaning also that update_history and general inquiry are finished, the command has not been generated due to CC update history mechanism (source is different than e_boot_update_pc and e_recover_update_pc).

- “receiving a command for an unknown element from CC”

SEA receives a command from CC for a not configured element. Note that the connection is fully established meaning also that update_history and general inquiry are finished, the command has not been generated due to CC update history mechanism (source is different than e_boot_update_pc and e_recover_update_pc).

- “SEA shall report the element state <CCState> with command reply flag <Flag> to CC”

SEA sends the signal element state <CCState> to CC via the s_epa_gcx_C20A_gen_elem_stat /s_epb_gcx_D30A_gen_elem_stat CHILL signal. Besides the element state itself it contains also <Flag>, indicating whether the state is reported as command reply or a spontaneous state change. This is represented as mode m_logical (true / false).

OC-IO Interface:

SCI-IO commands and messages referred in the following steps are defined in SCI-IO [14]. These steps are applicable for Station Keylock Single Lock, Line Keylock Server and Line Keylock Client.

- "changing the channels state reaching condition <Condition>"

Changing one or multiple input channels or checkback channel to reach an interested condition (set of states). For the case of multiple input channel changes, state change of several channels is received from the OC-IO within the same State of Input Channels message. These conditions are AllInputChannelsOff, AllInputChannelsNotOff, InputChannelsState_WITHDRAWN, InputChannelsState_not_WITHDRAWN, ChannelsState_LOCKED, ChannelsState_not_LOCKED as described below:

- AllInputChannelsOff: for KLS - this set (Key_Locked_in, Key_Released_in, Key_Supervision_in, Key_Withdrawn_in) = (OFF, OFF, OFF, OFF)
- AllInputChannelsNotOff: for KLS - any other set where the (Key_Locked_in, Key_Released_in, Key_Supervision_in, Key_Withdrawn_in) != (OFF, OFF, OFF, OFF)
- InputChannelsState_WITHDRAWN: for KLS - this set (Key_Locked_in, Key_Released_in, Key_Supervision_in, Key_Withdrawn_in) = (OFF, OFF, OFF, ON)
- InputChannelsState_not_WITHDRAWN: for KLS - any other set where the (Key_Locked_in, Key_Released_in, Key_Supervision_in, Key_Withdrawn_in) != (OFF, OFF, OFF, ON)
- ChannelsState_LOCKED: for KLL Server - this set (SLVFE_CB, SLFE_CB, M1_Key_Supervision_IN, M2_Ack_Button_IN) = (OFF, OFF, ON, OFF)

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- ChannelsState_not_LOCKED: for KLL Server - any other set where the (SLVFE_CB, SLFE_CB, M1_Key_Supervision_IN, M2_Ack_Button_IN) != (OFF, OFF, ON, OFF)
 - ChannelsState_LOCKING: for KLL Server - this set (SLVFE_CB, SLFE_CB, M1_Key_Supervision_IN, M2_Ack_Button_IN) = (OFF, OFF, OFF, OFF)
 - ChannelsState_not_LOCKED_not_LOCKING: for KLL Server - any other set where the (SLVFE_CB, SLFE_CB, M1_Key_Supervision_IN, M2_Ack_Button_IN) != {(OFF, OFF, ON, OFF); (OFF, OFF, OFF, OFF)}
- "when changing the input channels [<INPUT_CHANNEL_1>, <INPUT_CHANNEL_2>] to [<STATE_1>, <STATE_2>]"

State change of both channels is received from the OC-IO within the same State of Input Channels message.

- "when changing the input channels [<INPUT_CHANNEL_1>, <INPUT_CHANNEL_2>] to [<STATE_1>, <STATE_2>] within ****timer**** ms"

State change of both channels is received from the OC-IO within the same State of Input Channels message performed within specified time interval.

- "the current input channels state is <Condition>"

This step applicable for Keylock Single Lock (KLS) describes that all inputs channels states meet the defined condition, which can be one of AllInputChannelsOff, AllInputChannelsNotOff

- "the OC-IO state of channels evaluates to <Condition>"

This step describes that the OC-IO states of channels meet the defined condition, which can be one of AllInputChannelsOff, AllInputChannelsNotOff, ChannelsState_LOCKED, ChannelsState_LOCKING, ChannelsState_not_LOCKED_not_LOCKING.

- "the OC-IO state of the input channel <INPUT_CHANNEL> evaluates to <STATE>"

This step describes that the OC-IO state of the input channel meets the defined state.

- "the last stable element state is <LastStable>"

This step is used for Station Keylock - Single Lock when the last stable element state is relevant for processing commands from CC, input channels change, aspect diagnosis query from DGP or connection establishment with CC.

Current element state	Last stable element state	Current OC-IO state	Scenario
LOCKING	DISCONNECTED	DISCONNECTED	establishing the connection between ECMUX and OC-IO
LOCKING	DISTURBED	DISTURBED	one input channel state changes so all input channels state is OFF or timeout expiry while all input channels state is OFF
LOCKING	LOCKED /RELEASED /WITHDRAWN	LOCKED /RELEASED /WITHDRAWN	in PREPARE_LOCKED state, one input channel state changes so all input channels state is OFF or timeout expiry while all input channels state is OFF

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PREPARE_LOCKED	LOCKED /RELEASED	LOCKED /RELEASED	CC connection loss
PREPARE_LOCKED	RELEASED	RELEASED	cmd e_key_rel_canc_p executed
PREPARE_LOCKED	WITHDRAWN	WITHDRAWN	Key_Withdrawn_in input channel state change to OFF

Table 1: Details explaining the step *the last stable element state is LastStable*

Interface ECMUX - OC-IO connection:

- “losing the connection between ECMUX and OC-IO”:

Eulynx connection between the ECMUX and OC-IO is lost, ECMUX sets all elements controlled by this OC-IO to the default state and executes safe commands for all elements.

- “establishing the connection between ECMUX and OC-IO”:

Eulynx connection between the ECMUX and OC-IO is established, all phases required for an operational connection like version handshake and general inquiry are successfully completed.

- “SEA shall execute ECMUX safety reaction OC Disconnection with Close Reason *CloseReason*”:

This step includes sending “Close PDI” command to OC-IO, disconnect the connection between ECMUX and OC-IO, set element state to default state and report and diagnose state change and report and diagnose the disconnect. According with Interface specification SCI Generic the CloseReason values can be: PROTOCOL_ERROR(0x01), FORMAL_TELEGRAM_ERROR(0x02), CONTENT_TELEGRAM_ERROR(0x03), NORMAL_CLOSE(0x04), OTHER_VERSION_REQUIRED(0x05), TIMEOUT(0x06), CHECKSUM_MISMATCH(0x07).

Interface ECMUX - OC-IO:

- “receiving the SCI-IO message State Of Input Channels with State of channel x *ChannelState* from OC-IO”:

SEA receives the Message State Of Input Channels from OC-IO. Each *Input* defined in the “State of channel x” is forwarded to the SEA Input Channel configured to observe the SCI-IO Input Channel “x”. The received value can be: Channel Switched Off (0x01), Channel Switched On (0x02), Channel Disturbed (0x03).

- “receiving the SCI-IO message State Of Output Channels with Disturbance of channel x *ChannelState* from OC-IO”

SEA receives the Message State of Output Channels from OC-IO. Each *Output* defined in the “Disturbance of channel x” is forwarded to the SEA Output Channel configured to observe the SCI-IO Output Channel “x”. The received value can be: Channel Not Physically Disturbed (0x01), Channel Physically Disturbed (0x02).

- “SEA shall forward the command SCI-IO Set Output Channels with State of channel x *ChannelCmd* to OC-IO”

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SEA forwards the command SCI-IO Set Output Channels with State of channel x *ChannelCmd* to OC-IO. Each *Output* defined in the “State of channel x” is forwarded to the OC-IO configured to control the Output Channel “x”. The sent value can be: Channel Switched Off (0x01), Channel Switched On (0x02).

- “the current OC-IO state of input channel x is *OCState*”

In requirements SEA Channel Layer (see in SEA SW GEN SSS [5]) are referenced and not the OC-IO input channel directly. When the OC-IO is not connected, SEA does not have the information about the state of the OC-IO input channel and SEA input channel state cannot be directly reference as it is different from the real OC-IO input channel state. This expression defines the current state of the OC-IO input channel that shall be sent to ECMUX SEA Channel Layer during connection establishment procedure via Message State of Input Channels.

The OC-IO input channel x state can be: Channel Switched Off (0x01), Channel Switched On (0x02), Channel Disturbed (0x03).

- “the OC-IO state of the [input|output] channel x evaluates to *State*”

In requirements SEA Channel Layer (see in SEA SW GEN SSS [5]) input and output channel is referenced and not the OC-IO input/output channel directly. When the OC-IO is not connected, SEA does not have the information about the state of the OC-IO channels and SEA channel state cannot be directly reference as it is different from the real OC-IO input channel state. This expression defines the state that the SEA channel shall reach after OC-IO re-connection.

Possible values are: OFF, ON, DISTURBED.

- “the current OC-IO disturbance of output channel x is *OCState*”

In requirements SEA Channel Layer (see in SEA SW GEN SSS [5]) are referenced and not the OC-IO output channel directly. When the OC-IO is not connected, SEA does not have the information about the state of the OC-IO output channel and SEA output channel state cannot be directly reference as it is different from the real OC-IO output channel state. This expression defines the current state of the OC-IO output channel that shall be sent to ECMUX SEA Channel Layer during connection establishment procedure via Message State of Output Channels.

The OC-IO output channel x state can be: Channel Not Physically Disturbed (0x01), Channel Physically Disturbed (0x02).

- “SEA shall command the output channel x to OFF / ON”

This step consists of this series of actions:

- SEA shall forward the command SCI-IO Set Output Channels with State of channel CHANNEL_SWITCHED_OFF / CHANNEL_SWITCHED_ON to OC-IO
- And SEA shall set the current output channel state to OFF / ON
- And SEA shall report the current output channel state OFF / ON to OBSERVERS
- “OC-IO shall switch to safe state”

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When the OC-IO detects a disconnect of the ECMUX it shall command all output channels to OFF.

Interface SEA Channel Layer - OC-IO:

- “changing the *Channel* input channel to *State*”

SEA Channel Layer input channel receives a state update from the OC-IO via the SCI-IO Message State Of Input Channels and notifies the element process about the state change. For details on SEA Channel Layer (see in SEA SW GEN SSS [5]).

- “the current *Channel* input/output channel is *State*”

The SEA Channel Layer input channel corresponding to Channel can be ON / OFF / DISTURBED. The SEA Channel Layer monitored output channel corresponding to Channel can be ON / OFF / DISTURBED. The SEA Channel Layer not monitored output channel corresponding to Channel can be ON / OFF.

- “changing the *Channel* output channel to *State*”

SEA Channel Layer output channel receives a state update from the OC-IO via the SCI-IO message State Of Output Channels with Disturbance of channel and notifies the element process about the state change. For details on SEA Channel Layer see SEA SW GEN SSS [5]. The State can be ON / OFF / DISTURBED for monitored output channel and ON / OFF for not monitored output channel.

2.10 Abbreviations and Acronyms

For terms and abbreviations see [2].

2.11 Definitions and Terms

2.11.1 Terms

- “KLS” - Station Keylock Single Lock
- “KLL” - Line Keylock

2.11.2 Definitions

1. Configuration data
any information or file needed by the application software when executed (station data and software configuration files)
2. Software configuration files
files, which are loaded by the application at startup to read parameters for executing the software; not site specific

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3. Station data
the site specific preparation data, which describes the present elements of a component; output from Orest in XML format
4. Application software (files)
the executables (files), which are executed when the application is started
5. Element process
The ECMUX contains the logic corresponding to the behavior of each configured element, internally divided into a set of Java classes. This logic is referred to as “element process”.
6. Initial element state
The initial state is the state of the element upon startup when OC-IO is disconnected.
7. Default element state
The default state is the one the element shall try to recover to when an error occurs.
8. Keylocks - Station Keylock Single Lock, Line Keylock Server and Line Keylock Client
Keylock element is a special kind of point which is not equipped with electrical point machine, only with a key box. To switch this point, it is necessary to remove the key from the box. With this key, the point can be unlocked which enables to move the point manually. Station keylock is located in own station, line keylock server version is located on the line controlled by own station and line keylock client version is located on the line controlled by the neighboring station.

To describe parts of code, the following terms are used:

- Class
(usually) small part of executable code, encapsulating its own data in object-oriented context.
Example: “at.thales.sea.oec.builders.LampDeviceCfg”
- Package
group of classes contextually fitting together (e.g. depending on each other). Example: “at.thales.sea.oec.builders”. Grouping may be applied recursively; a package may contain sub-packages.
- Module
A Maven module. Consists of one or more packages. This is the smallest unit that can be built on its own. Example: “ecbase”.
- Component
Software part that serves a defined task. Consists of one or more modules. Note that modules may be used in multiple components, so in general components share the same modules.
Example: “ECMUX generic”. Note further, that this is not the definition of the term component as stated in CENELEC standard, but a term used in SW context.
- Deliverable
Software part that may be deployed on its own. This includes binaries, scripts, and configuration items like resource files or station data.

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3 Requirements

3.1 Keylock Element Processes

3.1.1 Line Keylocks

With the Line Keylock (German: Streckenschlüsselsperre), a key which is needed e.g. to switch a point on the line can be released to the operator of the point. When released, the key needed to switch the position of the point, can be taken out of the housing and inserted into the point to unlock it and switch its position. When the point is back in the normal position, the key can be removed from it and put back into the housing of the line keylock where it shall be locked again. All changes of the line keylock are reported to both the operating station and to the neighboring station.

The hardware, also shown in the image below, has a lock for the key as well as two lamps and two buttons that support the release process:

- Pre-release lamp located above the Acknowledge button (German: Vorfreigabe-Meldelampe): indicates that pre-release has been executed by the interlocking
- Acknowledge button (QT, German: Quittierungstaste): used to acknowledge the pre-release
- Release lamp located above the Key button (German: Freigabe-Meldelampe): indicates that the key has been released by the interlocking
- Key button (SLT, German: Schlüsseltaste): pressed to unlock the released key to take it out of the housing

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Figure 4: ECMUX Line Keylock - Real Hardware

Communication between ECMUX and HW is done via OC-IO (see EULYNX Requirements specification for Generic IO subsystem [12] and EULYNX Interface specification SCI-IO [14]). OC-IO channels are represented in ECMUX as channels of the ECMUX Channel Layer (see [5] for details).

The following ECMUX Channel Layer input channels are configured for both variants of Line keylocks:

Name	Functionality	ECMUX Channel Layer input channel type	OC-IO channel configuration
M1_Key_Supervision_IN	state of the supervision relay; OFF meaning that the key is unlocked and can be taken out	input channel (not safety relevant)	dual equivalent
M2_Ack_Button_IN	Acknowledge button supervision, OFF meaning that the button is not pressed	input channel (not safety relevant)	dual equivalent

Table 2: ECMUX Line keylock - Input Channels

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For specifics see following sub-chapters.

3.1.1.1 Line Keylock Server

The KLL server version is operated by its own interlocking and therefore can be commanded by the CC and can also command its HW.

Line keylock server version is realized in ECMUX via the component `:eulx_kll_server` as shown in the following figure.

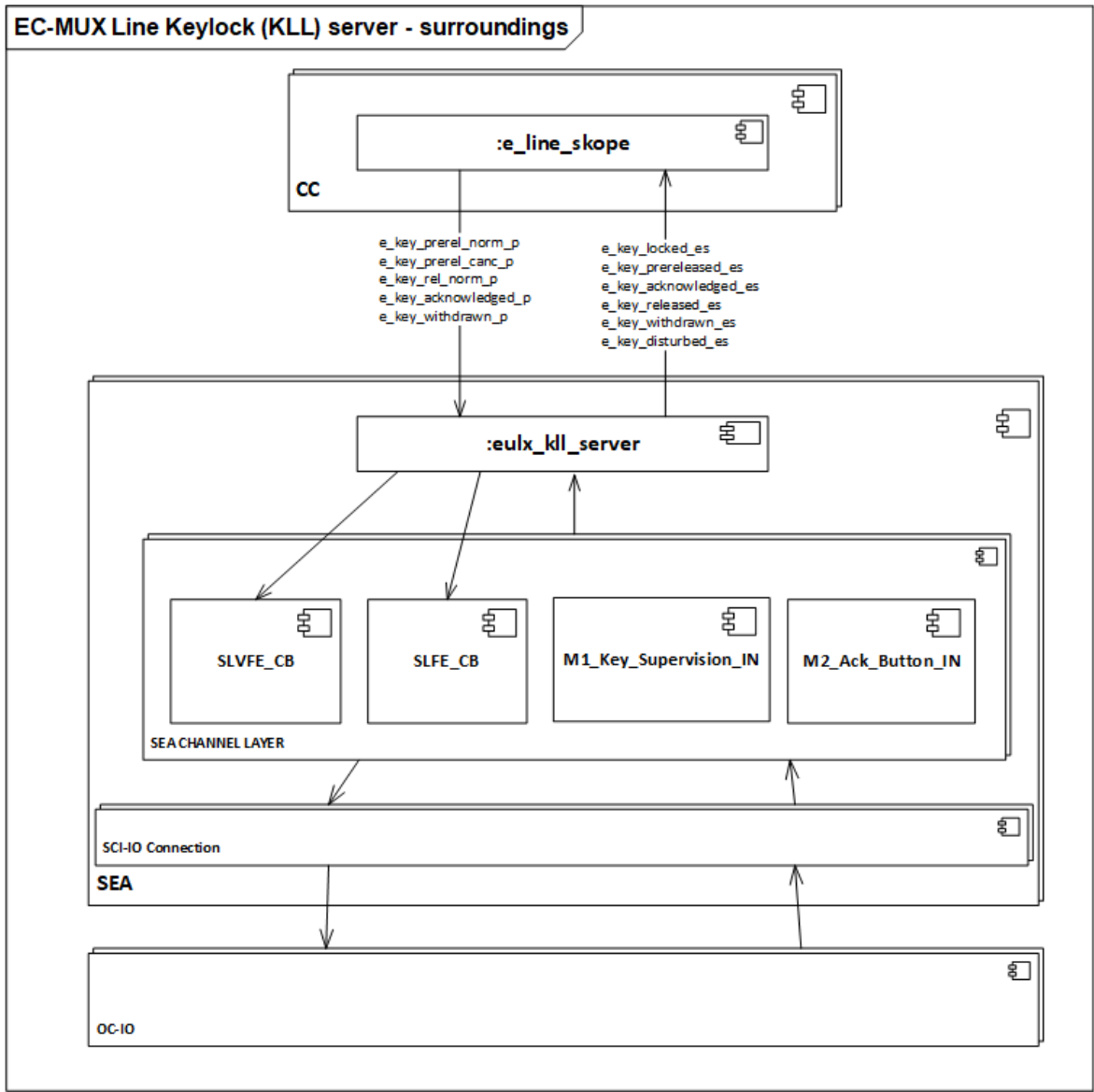


Figure 5: ECMUX Line Keylock (KLL) server - Surroundings

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States

Following internal element states are defined for the ECMUX Line Keylock server version:

Element State	Functionality	CC State	Faulty
DISCONNECTED	no connection to OC-IO / default state	e_key_disturbed_es	yes
DISTURBED	OC-IO connected but an error during acknowledging procedure detected (unexpected state change of a channel detected or HW reports an error for the channel)	e_key_disturbed_es	yes
LOCKED	key is locked within the HW housing	e_key_locked_es	no
PRE_RELEASED	pre-release has been commanded by the operator	e_key_prereleased_es	no
QT_PRESSED	a state change on one of the input channels was detected indicating that the Acknowledge button has been pressed, allow QT_SYNC_TIMER time for the second input change to be detected as well, pre-release executed, the key is locked within the HW housing	e_key_acknowledged_es	no
ACKNOWLEDGING	pressing of the Acknowledge button processed (both HW input channel states have inverted), storing of the pre-release acknowledgement in HW initiated, allow ACK_STORE_TIMER time for execution	e_key_acknowledged_es	no
ACKNOWLEDGED	pre-release acknowledgement successfully stored by HW	e_key_acknowledged_es	no
RELEASED	release has been commanded by the operator and successfully executed by the HW; the keylock is released, the key is locked within the HW housing	e_key_released_es	no
WITHDRAWN	Key button pressed, key is not locked and can be taken out, from this point the key removal cannot be detected while the Key button is being pressed, allow WD_TIMER time for the key withdrawal	e_key_withdrawn_es	no
WITHDRAW_RETRY	Key button accidentally released in WITHDRAWN state before the key was removed from the housing, allow the operator to retry the key withdrawal by pressing the Key button again within WD_RETRY_TIMER before commanding the HW to lock the key	e_key_released_es	no

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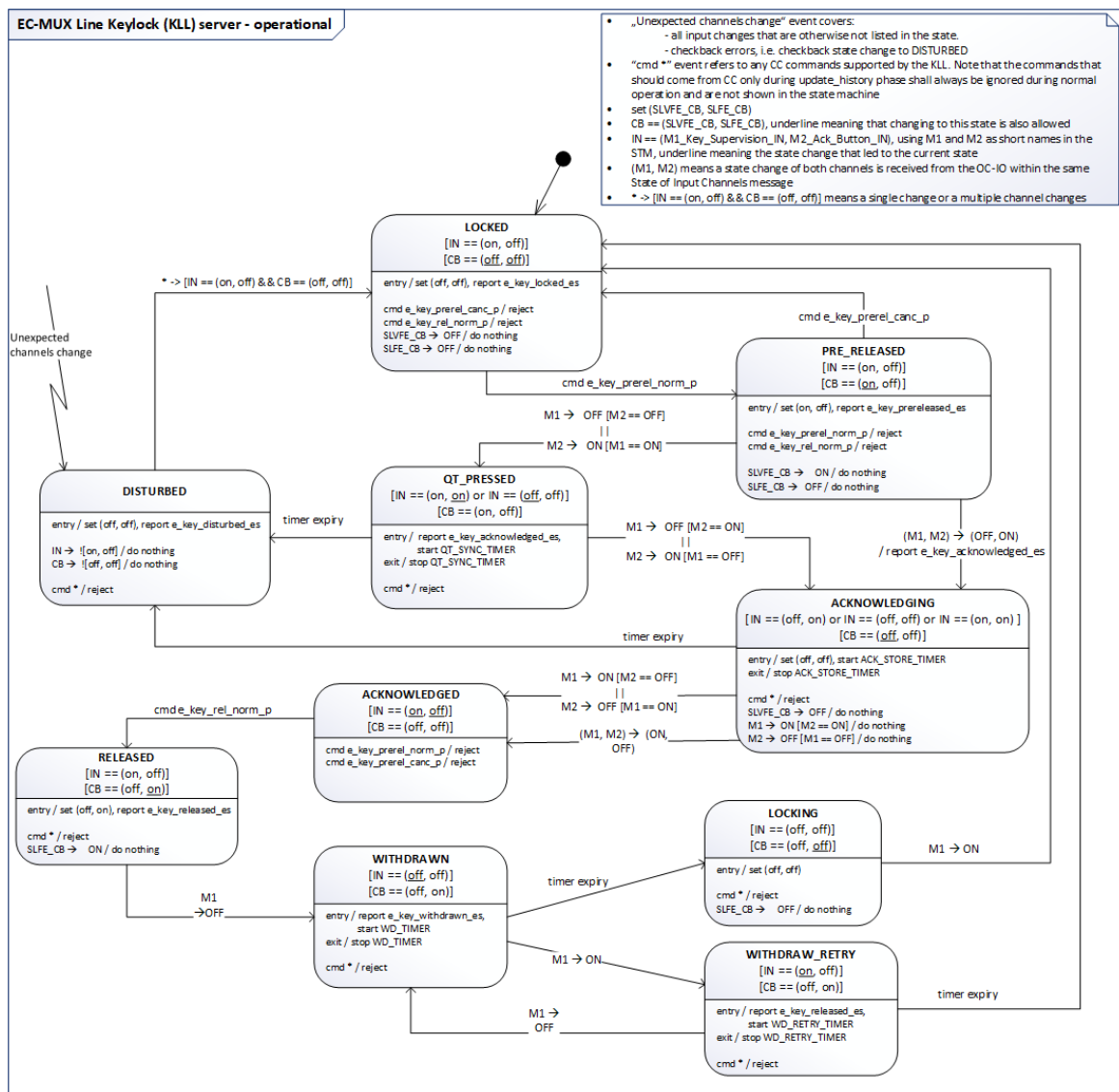
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LOCKING	the time allowed for the key withdrawal has run out, the key was either removed or the operator has not yet released the Key button, command to lock the key (when returned to the HW housing) has been executed, waiting for the key to be put back in the housing or for the operator to release the Key button if the key is in the housing	e_key_withdrawn_es	no
---------	--	--------------------	----

Table 3: ECMUX Line Keylock (KLL) server - Internal Element States

The state machine of the ECMUX Line keylock server version with operational scenarios is shown in the following figure:



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The state machine for the connection handling is shown in the following figure:



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As shown in the figure above, some of the internal states are timed in order to allow the HW to synchronise/execute commanded actions. Based on HW behavior, the following timer values are defined:

Name	Functionality	Default Value (milliseconds)
QT_SYNC_TIMER	Acknowledging timer - time for HW inputs M1_Key_Supervision_IN an M2_Ack_button_IN to synchronize after the Acknowledge button has been pressed. When the Acknowledge Button is pressed, the state might be detected and sent one by one by the OC-IO. Still, for the acknowledgement to be stored correctly in the HW, storing of the acknowledgement needs to be commanded within QT_SYNC_TIMER time from pressing the Acknowledge button.	800
ACK_STORE_TIMER	Acknowledgement storage timer - time for HW to successfully store the acknowledgement (time from commanding HW to store the acknowledgement until HW provides confirmation that it has been stored).	2000
WD_TIMER	Withdraw timer - time allowed for the key to be removed after the Key button was pressed.	30000
WD_RETRY_TIMER	Withdraw retry timer - time allowed for the Key button to be pressed again after it was presumably released by accident before the HW is commanded to lock the key.	60000
CB_TIMER	Check-back timer - Supervision of checkback functionality of the HW: When an output channel is commanded, successful execution of the command needs first to be confirmed via the corresponding input channel state change, within CB_TIMER time. Only afterward this, the HW shall react to the command by a subsequent change of one of the configured inputs (M1_Key_Supervision_IN or M2_Ack_Button_IN). If the input confirmation is not received within this time, KLL checkback channels SLVFE_CB and SLFE_CB shall switch to DISTURBED stated, causing KLL to also reach its DISTURBED state.	1500

Table 4: ECMUX Line Keylock (KLL) server - Timers

The default values for the timers listed above have been calculated based on the HW and OC-IO timings. The figure below shows the Acknowledge button pressed and the amount of time needed by the HW to transmit this information to ECMUX.

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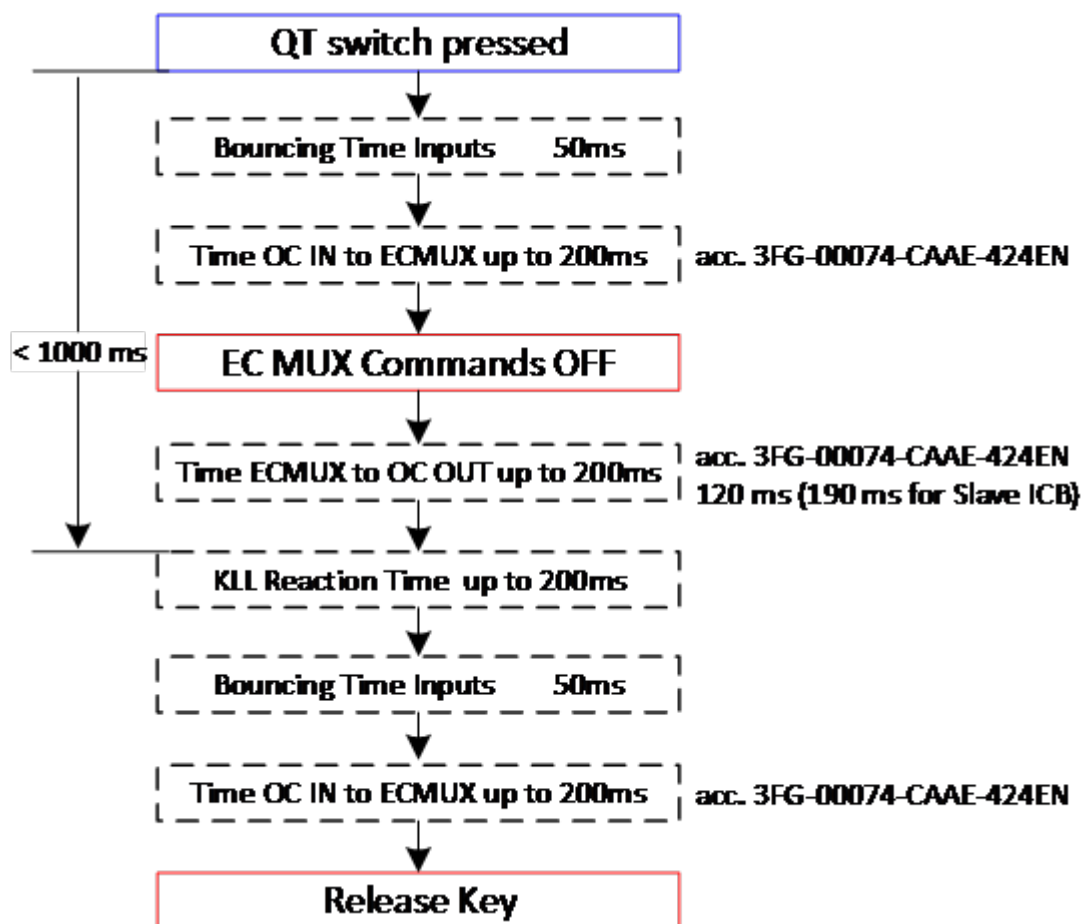


Figure 8: ECMUX Line Keylock (KLL) server - Acknowledge button pressed

All timer values can be configured to non-default values via ECMUX property files.

For additional information see [12].

HW Interface

In addition to the already listed input channels, the Line Keylock server version also has checkback channels (see [5] for details).

Name	Functionality	OC-IO Configuration
SLVFE_CB (German: Schlüssel-Vorfreigabe-Empfangsrelais)	Commands and confirms that command was executed for the pre-release of KLL when commanded to ON	2 dual valent OC-IO monitored output channels commanded valent, 1 single OC-IO input channel
SLFE_CB (German: Schlüssel-Freigabe-Empfangsrelais)	Commands and confirms that command was executed for the release of KLL when commanded to ON	2 dual valent OC-IO monitored output channels commanded valent, 1 single OC-IO input channel

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Table 5: ECMUX Line Keylock (KLL) server - Checkback Channels

CC Interface

The following CHILL signals for the line keylock (KLL) are used: s_gca_gpx_2C80_peri_out for channel A and s_gcb_gpx_3D80_peri_out for channel B. Note that the KLL supports execution of commands during normal operation and during connection establishment with CC (update_history phase). Command generated due to normal operation shall have command source *e_signal_man* or *e_automatic*, command generated due to update of history shall have command source *e_boot_update_pc* or *e_recover_update_pc*. For details about the connection establishment with CC and update_history phase see Chapter "CC connection establishment" in SSDD ECMUX GEN [3] and IDD CC/DGP - ECMUX [6].

The supported Line Keylock commands are listed below:

CC Command	Functionality	Processing phase
e_key_prerel_norm_p	Pre-release the line keylock or attempt to recover to the PRE_RELEASED state during update_history phase. When executed successfully, the pre-release lamp on HW shall light up.	operational, update_history
e_key_prerel_canc_p	Cancel the pre-release of the line keylock.	operational
e_key_rel_norm_p	Release the line keylock or attempt to recover to the RELEASED state during update_history phase. When executed successfully, the release lamp on HW shall light up.	operational, update_history
e_key_acknowledged_p	Attempt to recover the ACKNOWLEDGED state during update_history phase.	update_history
e_key_withdrawn_p	Attempt to recover the WITHDRAWN state during update_history phase.	update_history
e_key_locked_p	Not supported, shall cause a safety reaction ECMUX Deactivation.	none

Table 6: ECMUX Line Keylock (KLL) server - Commands (CC to ECMUX)

3.1.1.1.1 Operational - Line Keylock Server

In order to control and supervise a Line Keylock Server

As a SEA

I want to be able to translate input channels changes and CC commands into Line Keylock Server state changes and report them to CC

For requirement language see Chapter "Requirement Language".

Given SEA with a configured Line Keylock Server element

And the mapping between element state and checkback channels state is defined

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Element state	SLVFE_ CB	SLFE_ CB
DISTURBED	OFF	OFF
LOCKED	OFF	OFF
PRE_ RELEASED	ON	OFF
QT_ PRESSED	ON	OFF
ACKNOWLEDGING	OFF	OFF
ACKNOWLEDGED	OFF	OFF
RELEASED	OFF	ON
WITHDRAWN	OFF	ON
LOCKING	OFF	OFF
WITHDRAW_ RETRY	OFF	ON

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
KEY_ LOCKED	9
KEY_ DISTURBED	11
KEY_ PRE_ RELEASED	27
KEY_ ACKNOWLEDGED	29
KEY_ RELEASED	8
KEY_ WITHDRAWN	10

SEA-KLL:SSS:REQ: 4300	Line Keylock Server - Initial state		
When initializing the element			
Then SEA shall set the current element state to DISCONNECTED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory

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Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4301	Line Keylock Server - LOCKED - Pre-Rel command - switch to PRE_RELEASED		
Given the current element state is LOCKED When receiving the command e_key_prerel_norm_p from CC Then SEA shall switch to PRE_RELEASED And SEA shall report the element state e_key_prereleased_es with command reply flag true to CC And SEA shall report the diagnosis aspect message KEY_PRE_RELEASED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4302	Line Keylock Server - LOCKED - unexpected input channel changes - transition to DISTURBED
Given the current element state is LOCKED When changing the <InputChannel> input channel to <State> Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP	

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InputChannel	State
M1_Key_Supervision_IN	OFF
M1_Key_Supervision_IN	DISTURBED
M2_Ack_Button_IN	ON
M2_Ack_Button_IN	DISTURBED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ:4303	Line Keylock Server - LOCKED - checkback channel changes - do nothing					
Given the current element state is LOCKED When changing the <CheckbackChannel> checkback channel to OFF Then SEA shall do nothing <table><tr><th>CheckbackChannel</th></tr><tr><td>SLVFE_ CB</td></tr><tr><td>SLFE_ CB</td></tr></table>				CheckbackChannel	SLVFE_ CB	SLFE_ CB
CheckbackChannel						
SLVFE_ CB						
SLFE_ CB						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			

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Verification Method:	Test	Verification Step:	module test, subit
Comments:	The SLVFE_CB and SLFE_CB checkback channels are commanded to OFF when LOCKED is reached. If a checkback channel was previously in state ON, it is commanded to CHANGING_OFF state followed by the switch to OFF after the HW confirms that the command is executed.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4305	Line Keylock Server - DISTURBED - input channel changes - do nothing		
Given the current element state is DISTURBED			
When changing the channels state reaching condition ChannelsState_not_LOCKED			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	OC-IO message with one or multiple channels state changes is received so that the channel states (checkback and input channels)-(state) does not match (SLVFE_CB, SLFE_CB, M1_Key_Supervision_IN, M2_Ack_Button_IN) = (OFF, OFF, ON, OFF).		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4306	Line Keylock Server - DISTURBED - input channel changes - transition to LOCKED
Given the current element state is DISTURBED	
When changing the channels state reaching condition ChannelsState_LOCKED	

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Then SEA shall set the current element state to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag false to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	The switch between DISTURBED and LOCKED happens when a OC-IO message with one or multiple channel state changes is received so that the channel states (checkback and input channels)-(state) match (SLVFE_CB, SLFE_CB, M1_Key_Supervision_IN, M2_Ack_Button_IN) = (OFF, OFF, ON, OFF).		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4309	Line Keylock Server - PRE_RELEASED - SLVFE_CB / SLFE_CB checkback channel change - do nothing								
Given the current element state is PRE_RELEASED When changing the <Channel> checkback channel to <State> Then SEA shall do nothing <table><tr><th>Channel</th><th>State</th></tr><tr><td>SLVFE_ CB</td><td>ON</td></tr><tr><td>SLFE_ CB</td><td>OFF</td></tr></table>				Channel	State	SLVFE_ CB	ON	SLFE_ CB	OFF
Channel	State								
SLVFE_ CB	ON								
SLFE_ CB	OFF								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW ECMUX						
Verification Method:	Test	Verification Step:	module test, subit						

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Comments:	The SLVFE_CB checkback channel is commanded to ON when PRE_RELEASED is reached. The SLVFE_CB checkback channel was previously in state OFF and it is commanded to CHANGING_ON state followed by the switch to ON after the HW confirms that the command is executed.
Satisfies:	EL4-VIL-SSS-REQ-01521

SEA-KLL:SSS:REQ: 4310	Line Keylock Server - PRE_RELEASED - unexpected input channel changes - switch to DISTURBED					
Given the current element state is PRE_RELEASED When changing the <InputChannel> input channel to DISTURBED Then SEA shall switch to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><th>InputChannel</th></tr><tr><td>M1_ Key_ Supervision_ IN</td></tr><tr><td>M2_ Ack_ Button_ IN</td></tr></table>				InputChannel	M1_ Key_ Supervision_ IN	M2_ Ack_ Button_ IN
InputChannel						
M1_ Key_ Supervision_ IN						
M2_ Ack_ Button_ IN						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.					
Satisfies:	EL4-VIL-SSS-REQ-01521					

SEA-KLL:SSS:REQ: 4342	Line Keylock Server - PRE_RELEASED - switch to LOCKED
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Given the current element state is PRE_RELEASED When receiving the command e_key_prerelease_cancel from CC Then SEA shall switch to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag true to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ:4321	Line Keylock Server - PRE_RELEASED - switch to ACKNOWLEDGING		
Given the current element state is PRE_RELEASED When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [OFF, ON] Then SEA shall switch to ACKNOWLEDGING And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

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SEA-KLL:SSS:REQ: 4312	Line Keylock Server - PRE_RELEASED - transition to QT_PRESSED								
Given the current element state is PRE_RELEASED When changing the <InputChannel> input channel to <State> Then SEA shall set the current element state to QT_PRESSED And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP <table><tr><th>InputChannel</th><th>State</th></tr><tr><td>M1_Key_Supervision_IN</td><td>OFF</td></tr><tr><td>M2_Ack_Button_IN</td><td>ON</td></tr></table>				InputChannel	State	M1_Key_Supervision_IN	OFF	M2_Ack_Button_IN	ON
InputChannel	State								
M1_Key_Supervision_IN	OFF								
M2_Ack_Button_IN	ON								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW ECMUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:									
Satisfies:	EL4-VIL-SSS-REQ-01521								

SEA-KLL:SSS:REQ: 4313	Line Keylock Server - QT_PRESSED - QT_SYNC_TIMER expiry - switch to DISTURBED		
When the current state of the element has been QT_PRESSED for QT_SYNC_TIMER ms Then SEA shall switch to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory

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Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4314	Line Keylock Server - QT_PRESSED - unexpected input channel changes - switch to DISTURBED																							
Given the current element state is QT_PRESSED And the current M1_Key_Supervision_IN input channel is <State> And the current M2_Ack_Button_IN input channel is <State> When changing the <InputChannel> input channel to <NewState> Then SEA shall switch to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><th>State</th><th>InputChannel</th><th>NewState</th></tr><tr><td>ON</td><td>M1_Key_Supervision_IN</td><td>DISTURBED</td></tr><tr><td>ON</td><td>M2_Ack_Button_IN</td><td>OFF</td></tr><tr><td>ON</td><td>M2_Ack_Button_IN</td><td>DISTURBED</td></tr><tr><td>OFF</td><td>M1_Key_Supervision_IN</td><td>ON</td></tr><tr><td>OFF</td><td>M1_Key_Supervision_IN</td><td>DISTURBED</td></tr><tr><td>OFF</td><td>M2_Ack_Button_IN</td><td>DISTURBED</td></tr></table>				State	InputChannel	NewState	ON	M1_Key_Supervision_IN	DISTURBED	ON	M2_Ack_Button_IN	OFF	ON	M2_Ack_Button_IN	DISTURBED	OFF	M1_Key_Supervision_IN	ON	OFF	M1_Key_Supervision_IN	DISTURBED	OFF	M2_Ack_Button_IN	DISTURBED
State	InputChannel	NewState																						
ON	M1_Key_Supervision_IN	DISTURBED																						
ON	M2_Ack_Button_IN	OFF																						
ON	M2_Ack_Button_IN	DISTURBED																						
OFF	M1_Key_Supervision_IN	ON																						
OFF	M1_Key_Supervision_IN	DISTURBED																						
OFF	M2_Ack_Button_IN	DISTURBED																						
Type:	FUR - Functional Requirement																							
Status:	draft	Stability:	stable																					
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																					
Originator:	Technics (R&D)	Allocation:	SW ECMUX																					
Verification Method:	Test	Verification Step:	module test, subit																					

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Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.
Satisfies:	EL4-VIL-SSS-REQ-01521

SEA-KLL:SSS:REQ: 4316	Line Keylock Server - QT_PRESSED - input channel change - switch to ACKNOWLEDGING											
Given the current element state is QT_PRESSED And the current M1_Key_Supervision_IN input channel is <State> And the current M2_Ack_Button_IN input channel is <State> When changing the <InputChannel> input channel to <NewState> Then SEA shall switch to ACKNOWLEDGING <table><tr><th>State</th><th>InputChannel</th><th>NewState</th></tr><tr><td>ON</td><td>M1_Key_Supervision_IN</td><td>OFF</td></tr><tr><td>OFF</td><td>M2_Ack_Button_IN</td><td>ON</td></tr></table>				State	InputChannel	NewState	ON	M1_Key_Supervision_IN	OFF	OFF	M2_Ack_Button_IN	ON
State	InputChannel	NewState										
ON	M1_Key_Supervision_IN	OFF										
OFF	M2_Ack_Button_IN	ON										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW ECMUX									
Verification Method:	Test	Verification Step:	module test, subit									
Comments:												
Satisfies:	EL4-VIL-SSS-REQ-01521											

SEA-KLL:SSS:REQ: 4317	Line Keylock Server - ACKNOWLEDGING - QT_SYNC_TIMER expiry - transition to DISTURBED
When the current state of the element has been ACKNOWLEDGING for ACK_STORE_TIMER ms	

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Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4318	Line Keylock Server - ACKNOWLEDGING - input channel change - do nothing								
Given the current element state is ACKNOWLEDGING And the current M1_Key_Supervision_IN input channel is OFF And the current M2_Ack_Button_IN input channel is ON When changing the <InputChannel> input channel to <State> Then SEA shall do nothing <table><tr><th>InputChannel</th><th>State</th></tr><tr><td>M1_Key_Supervision_IN</td><td>ON</td></tr><tr><td>M2_Ack_Button_IN</td><td>OFF</td></tr></table>				InputChannel	State	M1_Key_Supervision_IN	ON	M2_Ack_Button_IN	OFF
InputChannel	State								
M1_Key_Supervision_IN	ON								
M2_Ack_Button_IN	OFF								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW ECMUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:									

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Satisfies:	EL4-VIL-SSS-REQ-01521
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SEA-KLL:SSS:REQ: 4319	Line Keylock Server - ACKNOWLEDGING - checkback channel change - do nothing		
Given the current element state is ACKNOWLEDGING			
When changing the SLVFE_CB checkback channel to OFF			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	The SLVFE_CB checkback channel is commanded to OFF when ACKNOWLEDGING is reached. The SLVFE_CB checkback channel was previously in state ON and it is commanded to CHANGING_OFF state followed by the switch to OFF after the HW confirms that the command is executed.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4320	Line Keylock Server - ACKNOWLEDGING - unexpected input channel changes - transition to DISTURBED
Given the current element state is ACKNOWLEDGING And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to DISTURBED Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP	

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SupervisionInState	AckButtonInState	InputChannel
OFF	ON	M1_ Key_ Supervision_ IN
OFF	ON	M2_ Ack_ Button_ IN
ON	ON	M1_ Key_ Supervision_ IN
ON	ON	M2_ Ack_ Button_ IN
OFF	OFF	M1_ Key_ Supervision_ IN
OFF	OFF	M2_ Ack_ Button_ IN

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4322	Line Keylock Server - ACKNOWLEDGING - input channel changes - transition to ACKNOWLEDGED		
Given the current element state is ACKNOWLEDGING And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to <State> Then SEA shall set the current element state to ACKNOWLEDGED			
SupervisionInState	AckButtonInState	InputChannel	State
ON	ON	M2_ Ack_ Button_ IN	OFF
OFF	OFF	M1_ Key_ Supervision_ IN	ON

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4324	Line Keylock Server - ACKNOWLEDGING - multiple input channel change - transition to ACKNOWLEDGED		
Given the current element state is ACKNOWLEDGING When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [ON, OFF] Then SEA shall set the current element state to ACKNOWLEDGED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4323	Line Keylock Server - ACKNOWLEDGED - unexpected input channel changes - transition to DISTURBED
Given the current element state is ACKNOWLEDGED	
When changing the <InputChannel> input channel to <State>	

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Then SEA shall set the current element state to DISTURBED
And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP

InputChannel	State
M1_Key_Supervision_IN	OFF
M1_Key_Supervision_IN	DISTURBED
M2_Ack_Button_IN	ON
M2_Ack_Button_IN	DISTURBED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ:4325	Line Keylock Server - ACKNOWLEDGED - Release norm command - switch to RELEASED		
Given the current element state is ACKNOWLEDGED When receiving the command e_key_rel_norm_p from CC Then SEA shall switch to RELEASED And SEA shall report the element state e_key_released_es with command reply flag true to CC And SEA shall report the diagnosis aspect message KEY_RELEASED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4327	Line Keylock Server - RELEASED - SLFE_CB checkback channel change - do nothing		
Given the current element state is RELEASED			
When changing the SLFE_CB checkback channel to ON			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	The SLFE_CB checkback channel is commanded to ON when RELEASED is reached. The SLFE_CB checkback channel was previously in state OFF and it is commanded to CHANGING_ON state followed by the switch to ON after the HW confirms that the command is executed.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4328	Line Keylock Server - RELEASED - unexpected input channel changes - switch to DISTURBED
Given the current element state is RELEASED When changing the <InputChannel> input channel to <State> Then SEA shall switch to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP	

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<table border="1"> <thead> <tr> <th>InputChannel</th><th colspan="3">State</th></tr> </thead> <tbody> <tr> <td>M1_Key_Supervision_IN</td><td colspan="3">DISTURBED</td></tr> <tr> <td>M2_Ack_Button_IN</td><td colspan="3">ON</td></tr> <tr> <td>M2_Ack_Button_IN</td><td colspan="3">DISTURBED</td></tr> </tbody> </table>				InputChannel	State			M1_Key_Supervision_IN	DISTURBED			M2_Ack_Button_IN	ON			M2_Ack_Button_IN	DISTURBED		
InputChannel	State																		
M1_Key_Supervision_IN	DISTURBED																		
M2_Ack_Button_IN	ON																		
M2_Ack_Button_IN	DISTURBED																		
Type:	FUR - Functional Requirement																		
Status:	draft	Stability:	stable																
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																
Originator:	Technics (R&D)	Allocation:	SW ECMUX																
Verification Method:	Test	Verification Step:	module test, subit																
Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.																		
Satisfies:	EL4-VIL-SSS-REQ-01521																		

SEA-KLL:SSS:REQ: 4330	Line Keylock Server - RELEASED - input channel changes - transition to WITHDRAWN		
Given the current element state is RELEASED When changing the M1_Key_Supervision_IN input channel to OFF Then SEA shall set the current element state to WITHDRAWN And SEA shall report the element state e_key_withdrawn_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_WITHDRAWN to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

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SEA-KLL:SSS:REQ:4331	Line Keylock Server - WITHDRAWN - WD_TIMER expiry - transition to LOCKING		
When the current state of the element has been WITHDRAWN for WD_TIMER ms			
Then SEA shall switch to LOCKING			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4332	Line Keylock Server - WITHDRAWN - unexpected input channel changes - switch to DISTURBED										
Given the current element state is WITHDRAWN When changing the <InputChannel> input channel to <State> Then SEA shall switch to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP											
<table><tr><th>InputChannel</th><th>State</th></tr><tr><td>M1_Key_Supervision_IN</td><td>DISTURBED</td></tr><tr><td>M2_Ack_Button_IN</td><td>ON</td></tr><tr><td>M2_Ack_Button_IN</td><td>DISTURBED</td></tr></table>				InputChannel	State	M1_Key_Supervision_IN	DISTURBED	M2_Ack_Button_IN	ON	M2_Ack_Button_IN	DISTURBED
InputChannel	State										
M1_Key_Supervision_IN	DISTURBED										
M2_Ack_Button_IN	ON										
M2_Ack_Button_IN	DISTURBED										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4334	Line Keylock Server - WITHDRAWN - input channel change - transition to WITHDRAW_RETRY		
Given the current element state is WITHDRAWN When changing the M1_Key_Supervision_IN input channel to ON Then SEA shall set the current element state to WITHDRAW_RETRY And SEA shall report the element state e_key_released_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_RELEASED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4307	Line Keylock Server - WITHDRAW_RETRY - WD_RETRY_TIMER expiry - transition to LOCKED
When the current state of the element has been WITHDRAW_RETRY for WD_RETRY_TIMER ms Then SEA shall switch to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4308	Line Keylock Server - WITHDRAW_RETRY - unexpected input channel changes - switch to DISTURBED										
Given the current element state is WITHDRAW_RETRY When changing the <InputChannel> input channel to <State> Then SEA shall switch to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><th>InputChannel</th><th>State</th></tr><tr><td>M1_ Key_ Supervision_ IN</td><td>DISTURBED</td></tr><tr><td>M2_ Ack_ Button_ IN</td><td>ON</td></tr><tr><td>M2_ Ack_ Button_ IN</td><td>DISTURBED</td></tr></table>				InputChannel	State	M1_ Key_ Supervision_ IN	DISTURBED	M2_ Ack_ Button_ IN	ON	M2_ Ack_ Button_ IN	DISTURBED
InputChannel	State										
M1_ Key_ Supervision_ IN	DISTURBED										
M2_ Ack_ Button_ IN	ON										
M2_ Ack_ Button_ IN	DISTURBED										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW ECMUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.										
Satisfies:	EL4-VIL-SSS-REQ-01521										

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SEA-KLL:SSS:REQ: 4315	Line Keylock Server - WITHDRAW_RETRY - input channel change - transition to WITHDRAWN		
Given the current element state is WITHDRAW_RETRY When changing the M1_Key_Supervision_IN input channel to OFF Then SEA shall set the current element state to WITHDRAWN And SEA shall report the element state e_key_withdrawn_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_WITHDRAWN to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4335	Line Keylock Server - LOCKING - SLFE_CB checkback channel change - do nothing		
Given the current element state is LOCKING When changing the SLFE_CB checkback channel to OFF Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	The SLFE_CB checkback channel is commanded to OFF when LOCKING is reached. The SLFE_CB checkback channel was previously in state ON and it is commanded to CHANGING_OFF state followed by the switch to OFF after the HW confirms that the command is executed.
Satisfies:	EL4-VIL-SSS-REQ-01521

SEA-KLL:SSS:REQ: 4336	Line Keylock Server - LOCKING - unexpected input channel changes - transition to DISTURBED										
Given the current element state is LOCKING When changing the <InputChannel> input channel to <State> Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><th>InputChannel</th><th>State</th></tr><tr><td>M1_Key_Supervision_IN</td><td>DISTURBED</td></tr><tr><td>M2_Ack_Button_IN</td><td>ON</td></tr><tr><td>M2_Ack_Button_IN</td><td>DISTURBED</td></tr></table>				InputChannel	State	M1_Key_Supervision_IN	DISTURBED	M2_Ack_Button_IN	ON	M2_Ack_Button_IN	DISTURBED
InputChannel	State										
M1_Key_Supervision_IN	DISTURBED										
M2_Ack_Button_IN	ON										
M2_Ack_Button_IN	DISTURBED										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW ECMUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:	Applicable for one or multiple channel changes in the same State of Input Channels message received from the OC-IO.										
Satisfies:	EL4-VIL-SSS-REQ-01521										

SEA-KLL:SSS:REQ:	Line Keylock Server - LOCKING - input channel changes - transition to

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4338	LOCKED		
Given the current element state is LOCKING When changing the M1_Key_Supervision_IN input channel to ON Then SEA shall set the current element state to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4337	Line Keylock Server - Reject commands	
Given the current element state is <State> When receiving the command <Command> from CC Then SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP		
State	Command	CCState
DISTURBED	e_key_prerel_norm_p	e_key_disturbed_es
DISTURBED	e_key_prerel_canc_p	e_key_disturbed_es
DISTURBED	e_key_rel_norm_p	e_key_disturbed_es
DISTURBED	e_key_acknowledged_p	e_key_disturbed_es
DISTURBED	e_key_withdrawn_p	e_key_disturbed_es
LOCKED	e_key_prerel_canc_p	e_key_locked_es
LOCKED	e_key_rel_norm_p	e_key_locked_es

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State	Command	CCState
LOCKED	e_key_acknowledged_p	e_key_locked_es
LOCKED	e_key_withdrawn_p	e_key_locked_es
PRE_RELEASED	e_key_prerel_norm_p	e_key_prereleased_es
PRE_RELEASED	e_key_rel_norm_p	e_key_prereleased_es
PRE_RELEASED	e_key_acknowledged_p	e_key_prereleased_es
PRE_RELEASED	e_key_withdrawn_p	e_key_prereleased_es
QT_PRESSED	e_key_prerel_norm_p	e_key_acknowledged_es
QT_PRESSED	e_key_prerel_canc_p	e_key_acknowledged_es
QT_PRESSED	e_key_rel_norm_p	e_key_acknowledged_es
QT_PRESSED	e_key_acknowledged_p	e_key_acknowledged_es
QT_PRESSED	e_key_withdrawn_p	e_key_acknowledged_es
ACKNOWLEDGING	e_key_prerel_norm_p	e_key_acknowledged_es
ACKNOWLEDGING	e_key_prerel_canc_p	e_key_acknowledged_es
ACKNOWLEDGING	e_key_rel_norm_p	e_key_acknowledged_es
ACKNOWLEDGING	e_key_acknowledged_p	e_key_acknowledged_es
ACKNOWLEDGING	e_key_withdrawn_p	e_key_acknowledged_es
ACKNOWLEDGED	e_key_prerel_norm_p	e_key_acknowledged_es
ACKNOWLEDGED	e_key_prerel_canc_p	e_key_acknowledged_es
ACKNOWLEDGED	e_key_acknowledged_p	e_key_acknowledged_es
ACKNOWLEDGED	e_key_withdrawn_p	e_key_acknowledged_es
RELEASED	e_key_prerel_norm_p	e_key_released_es

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State	Command	CCState
RELEASED	e_key_prerel_canc_p	e_key_released_es
RELEASED	e_key_rel_norm_p	e_key_released_es
RELEASED	e_key_acknowledged_p	e_key_released_es
RELEASED	e_key_withdrawn_p	e_key_released_es
WITHDRAWN	e_key_prerel_norm_p	e_key_withdrawn_es
WITHDRAWN	e_key_prerel_canc_p	e_key_withdrawn_es
WITHDRAWN	e_key_rel_norm_p	e_key_withdrawn_es
WITHDRAWN	e_key_acknowledged_p	e_key_withdrawn_es
WITHDRAWN	e_key_withdrawn_p	e_key_withdrawn_es
LOCKING	e_key_prerel_norm_p	e_key_withdrawn_es
LOCKING	e_key_prerel_canc_p	e_key_withdrawn_es
LOCKING	e_key_rel_norm_p	e_key_withdrawn_es
LOCKING	e_key_acknowledged_p	e_key_withdrawn_es
LOCKING	e_key_withdrawn_p	e_key_withdrawn_es
WITHDRAW_RETRY	e_key_prerel_norm_p	e_key_released_es
WITHDRAW_RETRY	e_key_prerel_canc_p	e_key_released_es
WITHDRAW_RETRY	e_key_rel_norm_p	e_key_released_es
WITHDRAW_RETRY	e_key_acknowledged_p	e_key_released_es
WITHDRAW_RETRY	e_key_withdrawn_p	e_key_released_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			

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Satisfies:	EL4-VIL-SSS-REQ-01521
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SEA-KLL:SSS:REQ:4304	Line Keylock Server - SLVFE_CB checkback channel changes - switch to DISTURBED												
Given the current element state is <State> When changing the SLVFE_CB checkback channel to DISTURBED Then SEA shall switch to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><th>State</th></tr><tr><td>LOCKED</td></tr><tr><td>PRE_ RELEASED</td></tr><tr><td>QT_ PRESSED</td></tr><tr><td>ACKNOWLEDGING</td></tr><tr><td>ACKNOWLEDGED</td></tr><tr><td>RELEASED</td></tr><tr><td>WITHDRAWN</td></tr><tr><td>LOCKING</td></tr><tr><td>WITHDRAW_ RETRY</td></tr></table>				State	LOCKED	PRE_ RELEASED	QT_ PRESSED	ACKNOWLEDGING	ACKNOWLEDGED	RELEASED	WITHDRAWN	LOCKING	WITHDRAW_ RETRY
State													
LOCKED													
PRE_ RELEASED													
QT_ PRESSED													
ACKNOWLEDGING													
ACKNOWLEDGED													
RELEASED													
WITHDRAWN													
LOCKING													
WITHDRAW_ RETRY													
Type:	FUR - Functional Requirement												
Status:	draft	Stability:	stable										
Target Release:	SEA SW GEN 3.0	Priority:	mandatory										
Originator:	Technics (R&D)	Allocation:	SW ECMUX										
Verification Method:	Test	Verification Step:	module test, subit										
Comments:													
Satisfies:	EL4-VIL-SSS-REQ-01521												

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SEA-KLL:SSS:REQ: 4311	Line Keylock Server - SLFE_CB checkback channel changes - switch to DISTURBED												
Given the current element state is <State> When changing the SLFE_CB checkback channel to DISTURBED Then SEA shall switch to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><th>State</th></tr><tr><td>LOCKED</td></tr><tr><td>PRE_ RELEASED</td></tr><tr><td>QT_ PRESSED</td></tr><tr><td>ACKNOWLEDGING</td></tr><tr><td>ACKNOWLEDGED</td></tr><tr><td>RELEASED</td></tr><tr><td>WITHDRAWN</td></tr><tr><td>LOCKING</td></tr><tr><td>WITHDRAW_ RETRY</td></tr></table>				State	LOCKED	PRE_ RELEASED	QT_ PRESSED	ACKNOWLEDGING	ACKNOWLEDGED	RELEASED	WITHDRAWN	LOCKING	WITHDRAW_ RETRY
State													
LOCKED													
PRE_ RELEASED													
QT_ PRESSED													
ACKNOWLEDGING													
ACKNOWLEDGED													
RELEASED													
WITHDRAWN													
LOCKING													
WITHDRAW_ RETRY													
Type:	FUR - Functional Requirement												
Status:	draft	Stability:	stable										
Target Release:	SEA SW GEN 3.0	Priority:	mandatory										
Originator:	Technics (R&D)	Allocation:	SW ECMUX										
Verification Method:	Test	Verification Step:	module test, subit										
Comments:													
Satisfies:	EL4-VIL-SSS-REQ-01521												

SEA-KLL:SSS:REQ: 4343	Line Keylock Server - Reboot when e_key_locked_p command is received
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Given the current element state is <State>

When receiving the command e_key_locked_p from CC

Then SEA shall execute ECMUX safety reaction Deactivation

State
DISCONNECTED
DISTURBED
LOCKED
PRE_ RELEASED
QT_ PRESSED
ACKNOWLEDGING
ACKNOWLEDGED
RELEASED
WITHDRAWN
LOCKING
WITHDRAW_ RETRY

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	The behavior is the same if the command is received during the history_phase for DISCONNECTED, DISTURBED, LOCKED, LOCKING, QT_PRESSED, ACKNOWLEDGING and ACKNOWLEDGED states.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

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3.1.1.1.2 Connection -Line Keylock Server

In order to control and supervise a Line Keylock Server

As a SEA

I want to be able to supervise connections and when changes occur then

- to execute commands during update_history phase, attempt to repair the commanded state and report it to CC
- on answering the general inquiry request from CC report the last stable element state to CC
- on CC connection loss switch to LOCKED for last stable element state PRE_RELEASED / RELEASED
- on OC-IO connection loss take on the default state DISCONNECTED
- on OC-IO connection establishment switch to DISTURBED state or transition to LOCKED state or transition to LOCKING state depending on the input channels state and checkback channels state
- report the connection state changes to CC and DGP

For requirement language see Chapter "Requirement Language".

Given SEA with a configured Line Keylock Server element

And the mapping between element state and checkback channels state is defined

Element state	SLVFE_ CB	SLFE_ CB
DISTURBED	OFF	OFF
LOCKED	OFF	OFF
PRE_ RELEASED	ON	OFF
QT_ PRESSED	ON	OFF
ACKNOWLEDGING	OFF	OFF
ACKNOWLEDGED	OFF	OFF
RELEASED	OFF	ON
WITHDRAWN	OFF	ON
LOCKING	OFF	OFF
WITHDRAW_ RETRY	OFF	ON

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
KEY_ LOCKED	9
KEY_ DISTURBED	11
KEY_ PRE_ RELEASED	27
KEY_	

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SEA diagnosis aspect	DGP diagnosis value
ACKNOWLEDGED	29
KEY_RELEASED	8
KEY_WITHDRAWN	10

SEA-KLL:SSS:REQ: 4470	Line Keylock Server - Execute command e_key_prerel_norm_p during update_history phase in LOCKED		
Given the current element state is LOCKED And CC connection state is UPDATE_HISTORY When receiving the command e_key_prerel_norm_p from CC Then SEA shall switch to PRE_RELEASED And SEA shall report the element state e_key_prereleased_es with command reply flag true to CC And SEA shall report the diagnosis aspect message KEY_PRE_RELEASED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4471	Line Keylock Server - Execute command e_key_acknowledged_p during update_history phase in LOCKED
Given the current element state is LOCKED And CC connection state is UPDATE_HISTORY When receiving the command e_key_acknowledged_p from CC Then SEA shall set the current element state to ACKNOWLEDGED And SEA shall report the element state e_key_acknowledged_es with command reply flag true to CC And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4472	Line Keylock Server - Execute command e_key_rel_norm_p during update_history phase in LOCKED		
Given the current element state is LOCKED And CC connection state is UPDATE_HISTORY When receiving the command e_key_rel_norm_p from CC Then SEA shall switch to RELEASED And SEA shall report the element state e_key_released_es with command reply flag true to CC And SEA shall report the diagnosis aspect message KEY_RELEASED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4473	Line Keylock Server - Ignore commands during update_history phase
Given the current element state is <State> And CC connection state is UPDATE_HISTORY	

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When receiving the command <Command> from CC

Then SEA shall do nothing

State	Command
LOCKED	e_key_withdrawn_p
LOCKED	e_key_prerel_canc_p
LOCKING	e_key_prerel_norm_p
LOCKING	e_key_prerel_canc_p
LOCKING	e_key_rel_norm_p
LOCKING	e_key_acknowledged_p
LOCKING	e_key_withdrawn_p
ACKNOWLEDGED	e_key_prerel_norm_p
ACKNOWLEDGED	e_key_prerel_canc_p
ACKNOWLEDGED	e_key_rel_norm_p
ACKNOWLEDGED	e_key_withdrawn_p
ACKNOWLEDGED	e_key_acknowledged_p
ACKNOWLEDGING	e_key_prerel_norm_p
ACKNOWLEDGING	e_key_prerel_canc_p
ACKNOWLEDGING	e_key_rel_norm_p
ACKNOWLEDGING	e_key_withdrawn_p
ACKNOWLEDGING	e_key_acknowledged_p
QT_PRESSED	e_key_prerel_norm_p
QT_PRESSED	e_key_prerel_canc_p
QT_PRESSED	e_key_rel_norm_p
QT_PRESSED	e_key_withdrawn_p
QT_PRESSED	e_key_acknowledged_p
DISTURBED	e_key_prerel_norm_p
DISTURBED	e_key_prerel_canc_p
DISTURBED	e_key_rel_norm_p
DISTURBED	e_key_withdrawn_p

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State	Command
DISTURBED	e_key_acknowledged_p
DISCONNECTED	e_key_prerel_norm_p
DISCONNECTED	e_key_prerel_canc_p
DISCONNECTED	e_key_rel_norm_p
DISCONNECTED	e_key_withdrawn_p
DISCONNECTED	e_key_acknowledged_p

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4401	Line Keylock Server - General Inquiry request from CC	
Given the current element state is <State> And CC connection state is GA_STARTED When answering the general inquiry request from CC Then SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall send the fault notification <SumFault> with fault type e_if_error to CC		
State	CCState	SumFault
DISCONNECTED	e_key_disturbed_es	sum_fault
DISTURBED	e_key_disturbed_es	sum_fault
LOCKED	e_key_locked_es	sum_fault_rep
		sum_fault_

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State	CCState	SumFault
PRE_RELEASED	e_key_prereleased_es	rep
QT_PRESSED	e_key_acknowledged_es	sum_fault_rep
ACKNOWLEDGING	e_key_acknowledged_es	sum_fault_rep
ACKNOWLEDGED	e_key_acknowledged_es	sum_fault_rep
RELEASED	e_key_released_es	sum_fault_rep
LOCKING	e_key_withdrawn_es	sum_fault_rep

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Note that before answering the general inquiry request, other stages of CC connection establishment are completed. States PRE_RELEASED and RELEASED are reached only due to execution commands during update_history phase. Other states are possible with or without execution of commands during update_history phase.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ:4402	Line Keylock Server - CC connection loss - do nothing
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall do nothing	

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State DISCONNECTED DISTURBED LOCKED QT_ PRESSED ACKNOWLEDGING ACKNOWLEDGED LOCKING				
	Type:	FUR - Functional Requirement		
	Status:	draft	Stability:	stable
	Target Release:	SEA SW GEN 3.0	Priority:	mandatory
	Originator:	Technics (R&D)	Allocation:	SW ECMUX
	Verification Method:	Test	Verification Step:	module test, subit
	Comments:			
	Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4403	Line Keylock Server - CC connection loss in PRE_RELEASED / RELEASED - switch to LOCKED				
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall switch to LOCKED And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP <table><tr><th>State</th></tr><tr><td>PRE_ RELEASED</td></tr><tr><td>RELEASED</td></tr><tr><td>WITHDRAW_ RETRY</td></tr></table>		State	PRE_ RELEASED	RELEASED	WITHDRAW_ RETRY
State					
PRE_ RELEASED					
RELEASED					
WITHDRAW_ RETRY					

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ:4409	Line Keylock Server - CC connection loss in WITHDRAWN - switch to LOCKING		
Given the current element state is WITHDRAWN And CC connection state is CONNECTED When losing the connection to CC Then SEA shall switch to LOCKING			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4404	Line Keylock Server - DISCONNECTED - [Re-]Connecting to OC-IO - transition to LOCKED
Given the current element state is DISCONNECTED And the OC-IO state of channels evaluates to ChannelsState_LOCKED When establishing the connection between ECMUX and OC-IO	

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Then SEA shall set the current element state to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag false to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4410	Line Keylock Server - DISCONNECTED - [Re-]Connecting to OC-IO - transition to LOCKING		
Given the current element state is DISCONNECTED And the OC-IO state of channels evaluates to ChannelsState_LOCKING When establishing the connection between ECMUX and OC-IO Then SEA shall set the current element state to LOCKING And SEA shall report the element state e_key_withdrawn_es with command reply flag false to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_WITHDRAWN to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

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SEA-KLL:SSS:REQ: 4405	Line Keylock Server - DISCONNECTED - [Re-]Connecting to OC-IO - transition to DISTURBED		
Given the current element state is DISCONNECTED And the OC-IO state of channels evaluates to ChannelsState_not_LOCKED_not_LOCKING When establishing the connection between ECMUX and OC-IO Then SEA shall set the current element state to DISTURBED And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4406	Line Keylock Server - Connection loss to OC-IO - DISTURBED		
Given the current element state is DISTURBED			
When losing the connection between ECMUX and OC-IO			
Then SEA shall set the current element state to DISCONNECTED			
And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

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SEA-KLL:SSS:REQ:4407	Line Keylock Server - Connection loss to OC-IO - transition to DISCONNECTED												
Given the current element state is <State> When losing the connection between ECMUX and OC-IO Then SEA shall set the current element state to DISCONNECTED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><th>State</th></tr><tr><td>LOCKED</td></tr><tr><td>PRE_ RELEASED</td></tr><tr><td>QT_ PRESSED</td></tr><tr><td>ACKNOWLEDGING</td></tr><tr><td>ACKNOWLEDGED</td></tr><tr><td>RELEASED</td></tr><tr><td>WITHDRAWN</td></tr><tr><td>LOCKING</td></tr><tr><td>WITHDRAW_ RETRY</td></tr></table>				State	LOCKED	PRE_ RELEASED	QT_ PRESSED	ACKNOWLEDGING	ACKNOWLEDGED	RELEASED	WITHDRAWN	LOCKING	WITHDRAW_ RETRY
State													
LOCKED													
PRE_ RELEASED													
QT_ PRESSED													
ACKNOWLEDGING													
ACKNOWLEDGED													
RELEASED													
WITHDRAWN													
LOCKING													
WITHDRAW_ RETRY													
Type:	FUR - Functional Requirement												
Status:	draft	Stability:	stable										
Target Release:	SEA SW GEN 3.0	Priority:	mandatory										
Originator:	Technics (R&D)	Allocation:	SW ECMUX										
Verification Method:	Test	Verification Step:	module test, subit										
Comments:													
Satisfies:	EL4-VIL-SSS-REQ-01521												

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SEA-KLL:SSS:REQ: 4408	Line Keylock Server - DISCONNECTED - reject commands - OC-IO NOT_CONNECTED								
Given the current element state is DISCONNECTED When receiving the command <Command> from CC Then SEA shall report the element state e_key_disturbed_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_key_prerel_norm_p</td></tr><tr><td>e_key_prerel_canc_p</td></tr><tr><td>e_key_rel_norm_p</td></tr><tr><td>e_key_acknowledged_p</td></tr><tr><td>e_key_withdrawn_p</td></tr></table>				Command	e_key_prerel_norm_p	e_key_prerel_canc_p	e_key_rel_norm_p	e_key_acknowledged_p	e_key_withdrawn_p
Command									
e_key_prerel_norm_p									
e_key_prerel_canc_p									
e_key_rel_norm_p									
e_key_acknowledged_p									
e_key_withdrawn_p									
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW ECMUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:									
Satisfies:	EL4-VIL-SSS-REQ-01521								

3.1.1.1.3 Diagnosis - Line Keylock Server

In order to provide the aspect diagnosis for a Line Keylock Server

As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried element

For requirement language see Chapter "Requirement Language".

Given SEA with a configured Line Keylock Server element

And the mapping between element state and checkback channels state is defined

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Element state	SLVFE_ CB	SLFE_ CB
DISTURBED	OFF	OFF
LOCKED	OFF	OFF
PRE_ RELEASED	ON	OFF
QT_ PRESSED	ON	OFF
ACKNOWLEDGING	OFF	OFF
ACKNOWLEDGED	OFF	OFF
RELEASED	OFF	ON
WITHDRAWN	OFF	ON
LOCKING	OFF	OFF
WITHDRAW_ RETRY	OFF	ON

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
KEY_ LOCKED	9
KEY_ DISTURBED	11
KEY_ PRE_ RELEASED	27
KEY_ ACKNOWLEDGED	29
KEY_ RELEASED	8
KEY_ WITHDRAWN	10

SEA-KLL:SSS:REQ: 4450	Line Keylock Server - aspect query - OC-IO CONNECTED
Given the current element state is <State> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP	

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State	DGPMessage
DISTURBED	KEY_ DISTURBED
LOCKED	KEY_ LOCKED
PRE_ RELEASED	KEY_ PRE_ RELEASED
QT_ PRESSED	KEY_ ACKNOWLEDGED
ACKNOWLEDGING	KEY_ ACKNOWLEDGED
ACKNOWLEDGED	KEY_ ACKNOWLEDGED
RELEASED	KEY_ RELEASED
WITHDRAW_ RETRY	KEY_ RELEASED
WITHDRAWN	KEY_ WITHDRAWN
LOCKING	KEY_ WITHDRAWN

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

SEA-KLL:SSS:REQ: 4452	Line Keylock Server - aspect query - DISCONNECTED - OC-IO NOT_CONNECTED
Given the current element state is DISCONNECTED When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP	
Type:	FUR - Functional Requirement

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.1.2 Line Keylock Client

The KLL client version is not operated by its own interlocking, but by the neighboring interlocking system. Therefore it cannot command the KLL HW, it reads in the states from the KLL HW as information for the own interlocking, and has limited support for executing CC commands.

Line keylock client version is realized in ECMUX via the component **:eulx_kll_client** as shown in the following figure.

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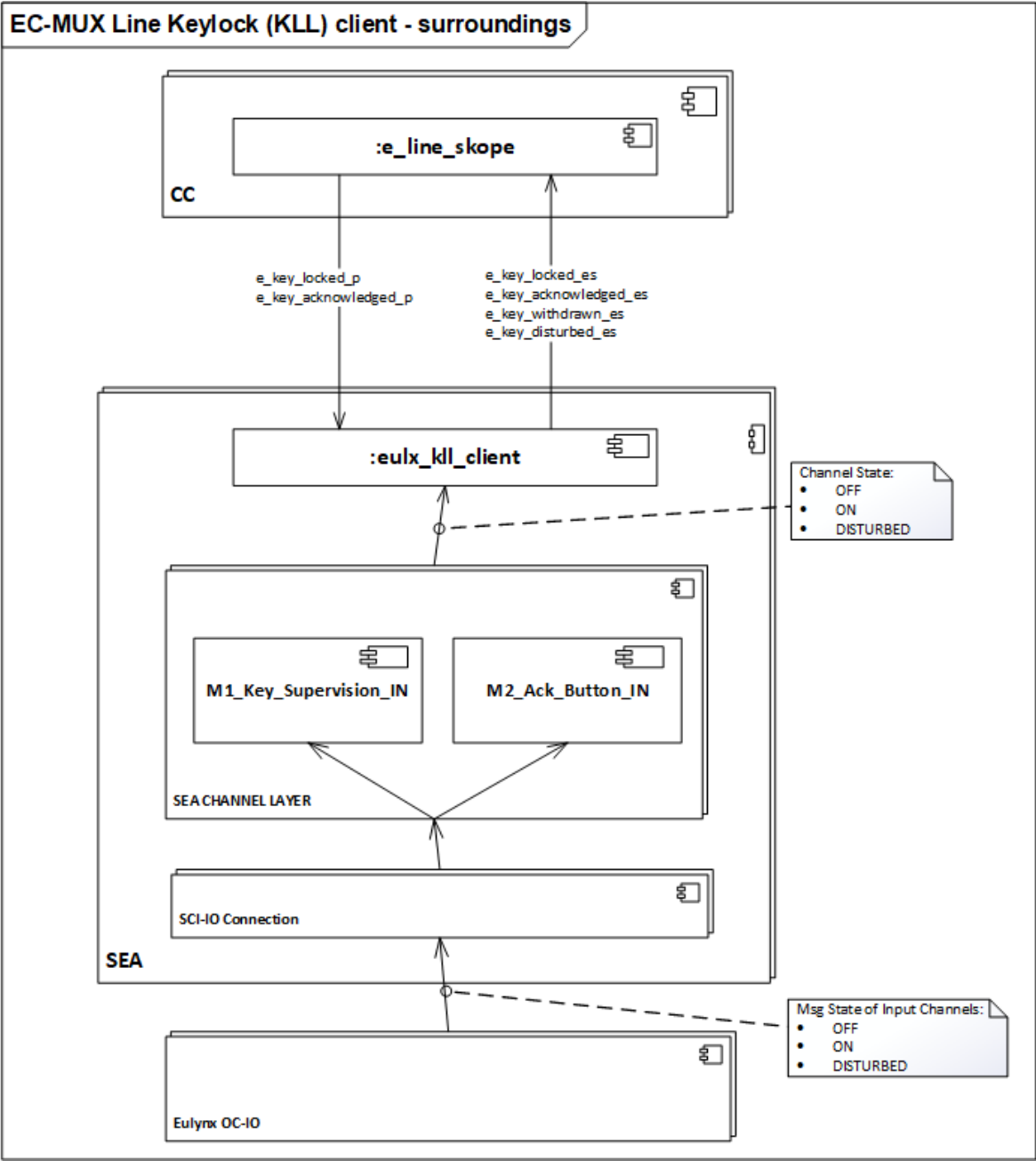


Figure 9: ECMUX Line Keylock client version - Surroundings

Following internal element states are defined for the ECMUX KLL client version:

Element State	Functionality	CC State	Faulty
---------------	---------------	----------	--------

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DISCONNECTED	no connection to OC-IO / default state	e_key_disturbed_es	yes
DISTURBED	OC-IO connected but an error was detected (unexpected state change of a channel detected or HW reports an error for the channel)	e_key_disturbed_es	yes
LOCKED	key is locked within the HW housing	e_key_locked_es	no
QT_PRESSED	state change on one of the inputs indicating that the Acknowledge button has been pressed, allow QT_SYNC_TIMER time for the second input change to be detected as well	e_key_locked_es	no
ACKNOWLEDGING	pressing of the Acknowledge button processed (both HW inputs inverted), storing of the acknowledgement in HW initiated, allow ACK_STORE_TIMER time for execution	e_key_acknowledged_es	no
ACKNOWLEDGED	pre-release confirmed, acknowledgement successfully stored by HW	e_key_acknowledged_es	no
WITHDRAWING	state change on one of the inputs indicating that the Key Button has been pressed, give the input channels a chance (SYNC_TIMER time) to stabilize into the next stable state [OFF OFF] corresponding to WITHDRAWN	e_key_acknowledged_es	no
WAIT_FOR_LOCK	after executing the command e_key_locked_p in state WITHDRAWING and immediately reporting the state e_key_locked_es, wait for input channels (SYNC_TIMER time) to stabilize into the next stable state [ON OFF] corresponding to reaching LOCKED	e_key_locked_es	no
WITHDRAWN	pressing of the Key button processed, from this point it cannot be guaranteed that the key has not been removed	e_key_withdrawn_es	no
LOCKING	Waiting for the key to be returned to the housing or for the input channels to be stable for a given time (SYNC_TIMER) before transitioning to LOCKED	e_key_withdrawn_es	no

Table 7: ECMUX Line keylock client version - Internal Element States

Because input channels have the same state combination in LOCKED and ACKNOWLEDGED, it is possible that while handling error scenarios the element process reaches state LOCKED instead of ACKNOWLEDGED. To handle these cases, CC implements following additional functionality:

- additional property for state LOCKED indicating if this state is trustworthy or not (ger. die Tauglichkeit der Grundstellung). LOCKED state is marked as not trustworthy when there is a

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possibility that the HW is really in state ACKNOWLEDGED. For more details see Chapter “4.1.3 Error Scenarios” in Software Design Description document ECMUX SW Keylock / EULYNX [4] and Chapter 4.16 in [11].

- additional logic that allows corrections of the current element state by executing the e_key_acknowledged_p or e_key_locked_p commands

The state machine of the ECMUX Line keylock client version with operational scenarios is shown in the following figure:

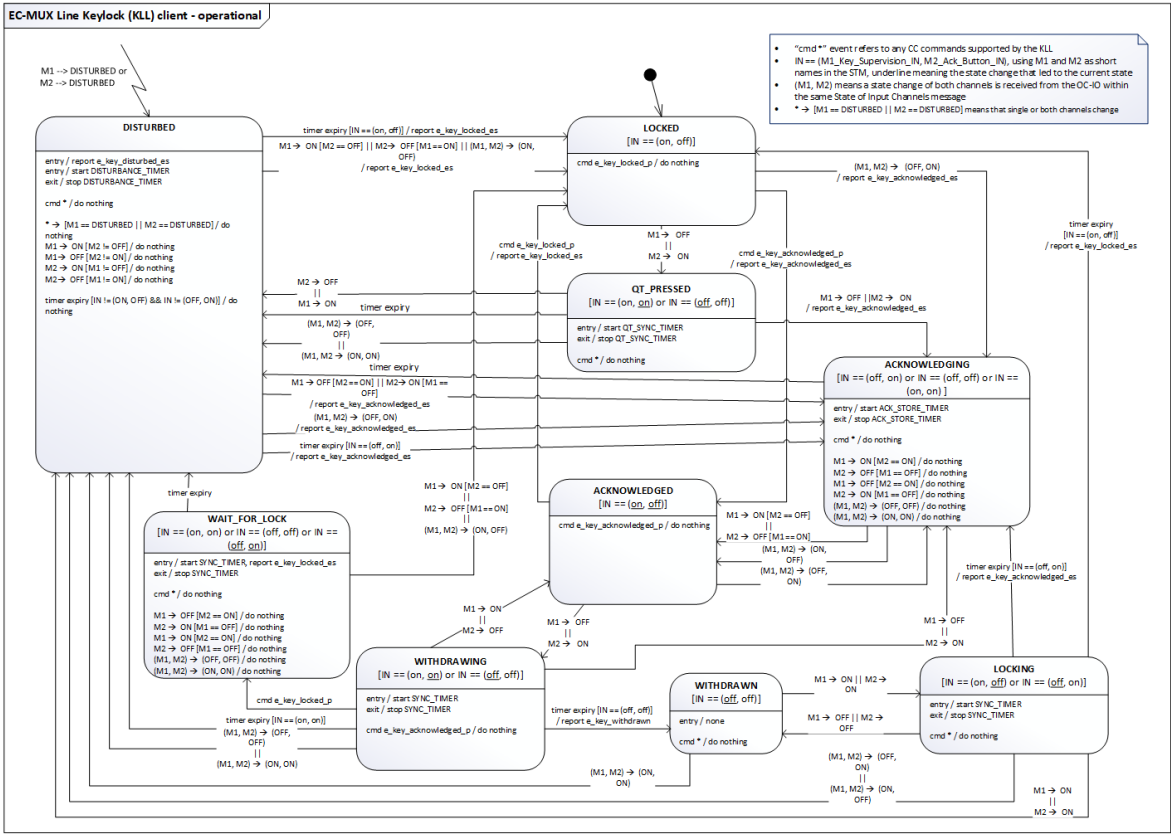


Figure 10: ECMUX Line keylock client version - State Machine

The state machine for the connection handling is shown in the following figure:

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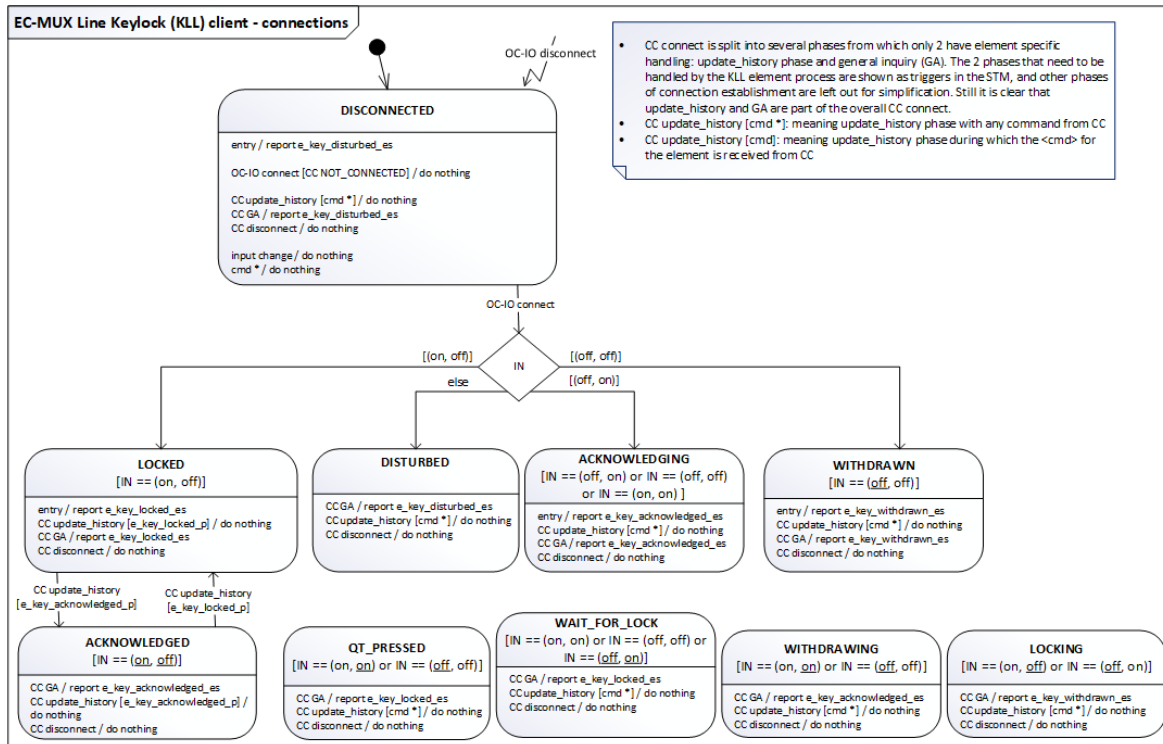


Figure 11: ECMUX Line Keylock (KLL) client - connection handling - State Machine

Name	Functionality	Value (milliseconds)
QT_SYNC_TIMER	Acknowledging timer - time to wait for the inputs to synch after the Acknowledge button has been pressed	1000
ACK_STORE_TIMER	Invert timer - for HW to successfully store the acknowledgement (time from commanding the HW to store the acknowledgement until HW confirmation that it is stored).	6000
SYNC_TIMER	time for the input channels states to be stable enough before allowing to reach the next state	1000
DISTURBANCE_TIMER	time to stay in DISTURBED before transitioning to LOCKED / ACKNOWLEDGING if the channels still match the states corresponding to the state LOCKED / ACKNOWLEDGING	2000

Table 8: ECMUX Line keylock client version - Timers

Timer values have been taken as in Functional Specification Line Keylocks [10]

HW Interface

Nothing additional to the already listed input channels.

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CC Interface

The following commands are generated during boot/re-connect scenario or operation when CC concludes that the keylock state has changed. E.g. Operator in the neighbor station canceled the pre-release which is not visible on the HW interface. If the block direction is then changed, CC knows that the line keylock already transitioned from ACKNOWLEDGED to LOCKED state and shall generate the e_key_locked_p command to bring own keylock element process to the same state.

CC Command	Functionality
e_key_locked_p	Key locked within the HW housing.
e_key_acknowledged_p	Pre-release acknowledged and stored in HW.

Table 9: ECMUX Line keylock client version - Commands (CC to ECMUX)

Note that e_key_prerelease_norm_p/e_key_prerelease_cancel_p/e_key_release_norm_p/e_key_withdrawn_p commands are not supported and shall cause a safety reaction ECMUX Deactivation.

3.1.1.2.1 Operational - Line Keylock - Client

In order to control and supervise a Line Keylock Client

As a SEA

I want to be able to translate input channels changes and CC commands into Line Keylock Client state changes and report them to CC

For requirement language see Chapter "Requirement Language".

Given SEA with a configured Line Keylock Client element

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
KEY_LOCKED	9
KEY_DISTURBED	11
KEY_ACKNOWLEDGED	29
KEY_WITHDRAWN	10

SEA-KLL:SSS:REQ: 4500	Line Keylock Client - Initial state
When initializing the element	

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Then SEA shall set the current element state to DISCONNECTED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4501	Line Keylock Client - LOCKED - Acknowledged command - transition to ACKNOWLEDGED		
Given the current element state is LOCKED			
When receiving the command e_key_acknowledged_p from CC			
Then SEA shall set the current element state to ACKNOWLEDGED			
And SEA shall report the element state e_key_acknowledged_es with command reply flag true to CC			
And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4502	Line Keylock Client - LOCKED - input channel changes - transition to QT_PRESSED
Given the current element state is LOCKED	

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When changing the <InputChannel> input channel to <State>

Then SEA shall set the current element state to QT_PRESSED

InputChannel	State
M1_ Key_ Supervision_ IN	OFF
M2_ Ack_ Button_ IN	ON

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4535	Line Keylock Client - LOCKED - input channel changes - transition to ACKNOWLEDGING		
Given the current element state is LOCKED When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [OFF, ON] Then SEA shall set the current element state to ACKNOWLEDGING And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

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SEA-KLL:SSS:REQ:4504

Line Keylock Client - DISTURBED - input channel changes to ON/OFF - do nothing

Given the current element state is DISTURBED
And the current M1_Key_Supervision_IN input channel is <SupervisionInState>
And the current M2_Ack_Button_IN input channel is <AckButtonInState>

When changing the <InputChannel> input channel to <State>

Then SEA shall do nothing

SupervisionInState	AckButtonInState	InputChannel	State
DISTURBED	DISTURBED	M1_Key_Supervision_IN	ON
OFF	DISTURBED	M1_Key_Supervision_IN	ON
DISTURBED	ON	M1_Key_Supervision_IN	ON
OFF	ON	M1_Key_Supervision_IN	ON
DISTURBED	DISTURBED	M1_Key_Supervision_IN	OFF
ON	DISTURBED	M1_Key_Supervision_IN	OFF
DISTURBED	OFF	M1_Key_Supervision_IN	OFF
ON	OFF	M1_Key_Supervision_IN	OFF
DISTURBED	DISTURBED	M2_Ack_Button_IN	ON
DISTURBED	OFF	M2_Ack_Button_IN	ON
ON	DISTURBED	M2_Ack_Button_IN	ON
ON	OFF	M2_Ack_Button_IN	ON
DISTURBED	DISTURBED	M2_Ack_Button_IN	OFF
DISTURBED	ON	M2_Ack_Button_IN	OFF
OFF	DISTURBED	M2_Ack_Button_IN	OFF
OFF	ON	M2_Ack_Button_IN	OFF

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	This behaviour is applicable whether the DISTURBANCE_TIMER is running / not running. One channel state changes so the final state combinations for M1_Key_Supervision_IN and M2_Ack_Button_IN doesn't match (ON, OFF) or (OFF, ON)		
Satisfies:	ELO-BF:FUR:2269		

SEA-KLL:SSS:REQ: 4505	Line Keylock Client - DISTURBED - input channel changes to DISTURBED - do nothing					
Given the current element state is DISTURBED When changing the <InputChannel> input channel to DISTURBED Then SEA shall do nothing <table><tr><td>InputChannel</td></tr><tr><td>M1_ Key_ Supervision_ IN</td></tr><tr><td>M2_ Ack_ Button_ IN</td></tr></table>				InputChannel	M1_ Key_ Supervision_ IN	M2_ Ack_ Button_ IN
InputChannel						
M1_ Key_ Supervision_ IN						
M2_ Ack_ Button_ IN						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	This behaviour is applicable whether the DISTURBANCE_TIMER is running / not running. Applicable for single or both input channels change to DISTURBED as reported by OC-IO in separate or the same SCI-IO State Input Channels message.					

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Satisfies:	EL4-VIL-SSS-REQ-01521
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SEA-KLL:SSS:REQ:4503	Line Keylock Client - DISTURBED - DISTURBANCE_TIMER expiry - do nothing																		
Given the current element state is DISTURBED And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When the current state of the element has been DISTURBED for DISTURBANCE_TIMER ms Then SEA shall do nothing <table><tr><th>SupervisionInState</th><th>AckButtonInState</th></tr><tr><td>ON</td><td>ON</td></tr><tr><td>OFF</td><td>OFF</td></tr><tr><td>ON</td><td>DISTURBED</td></tr><tr><td>OFF</td><td>DISTURBED</td></tr><tr><td>DISTURBED</td><td>ON</td></tr><tr><td>DISTURBED</td><td>OFF</td></tr><tr><td>DISTURBED</td><td>DISTURBED</td></tr></table>				SupervisionInState	AckButtonInState	ON	ON	OFF	OFF	ON	DISTURBED	OFF	DISTURBED	DISTURBED	ON	DISTURBED	OFF	DISTURBED	DISTURBED
SupervisionInState	AckButtonInState																		
ON	ON																		
OFF	OFF																		
ON	DISTURBED																		
OFF	DISTURBED																		
DISTURBED	ON																		
DISTURBED	OFF																		
DISTURBED	DISTURBED																		
Type:	FUR - Functional Requirement																		
Status:	draft	Stability:	stable																
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																
Originator:	Technics (R&D)	Allocation:	SW ECMUX																
Verification Method:	Test	Verification Step:	module test, subit																
Comments:																			
Satisfies:	ELO-BF:FUR:2269																		

SEA-KLL:SSS:REQ: 4506	Line Keylock Client - DISTURBED - input channel change - transition to LOCKED
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Given the current element state is DISTURBED
And the current M1_Key_Supervision_IN input channel is <SupervisionInState>
And the current M2_Ack_Button_IN input channel is <AckButtonInState>

When changing the <InputChannel> input channel to <State>

Then SEA shall set the current element state to LOCKED
And SEA shall report the element state e_key_locked_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP

SupervisionInState	AckButtonInState	InputChannel	State
DISTURBED	OFF	M1_Key_Supervision_IN	ON
OFF	OFF	M1_Key_Supervision_IN	ON
ON	DISTURBED	M2_Ack_Button_IN	OFF
ON	ON	M2_Ack_Button_IN	OFF

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	This behaviour is applicable whether the DISTURBANCE_TIMER is running / not running.		
Satisfies:	ELO-BF:FUR:2190, ELO-BF:FUR:2267		

SEA-KLL:SSS:REQ: 4509	Line Keylock Client - DISTURBED - input channel change - transition to ACKNOWLEDGING
Given the current element state is DISTURBED And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to <State>	

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Then SEA shall set the current element state to ACKNOWLEDGING
And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP

SupervisionInState	AckButtonInState	InputChannel	State
DISTURBED	ON	M1_Key_Supervision_IN	OFF
ON	ON	M1_Key_Supervision_IN	OFF
OFF	DISTURBED	M2_Ack_Button_IN	ON
OFF	OFF	M2_Ack_Button_IN	ON

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	This behaviour is applicable whether the DISTURBANCE_TIMER is running / not running.		
Satisfies:	ELO-BF:FUR:2191, ELO-BF:FUR:2268		

SEA-KLL:SSS:REQ: 4539	Line Keylock Client - DISTURBED - input channel changes - transition to LOCKED
Given the current element state is DISTURBED And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [ON, OFF] Then SEA shall set the current element state to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag false to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP	

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<table><tr><th>SupervisionInState</th><th>AckButtonInState</th></tr><tr><td>DISTURBED</td><td>DISTURBED</td></tr><tr><td>DISTURBED</td><td>ON</td></tr><tr><td>OFF</td><td>DISTURBED</td></tr></table>				SupervisionInState	AckButtonInState	DISTURBED	DISTURBED	DISTURBED	ON	OFF	DISTURBED
SupervisionInState	AckButtonInState										
DISTURBED	DISTURBED										
DISTURBED	ON										
OFF	DISTURBED										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW ECMUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:	This behaviour is applicable while the DISTURBANCE_TIMER is running / not running.										
Satisfies:	ELO-BF:FUR:2190, ELO-BF:FUR:2267										

SEA-KLL:SSS:REQ: 4507	Line Keylock Client - DISTURBED - DISTURBANCE_TIMER expiry - transition to LOCKED		
Given the current element state is DISTURBED And the current M1_Key_Supervision_IN input channel is ON And the current M2_Ack_Button_IN input channel is OFF When the current state of the element has been DISTURBED for DISTURBANCE_TIMER ms Then SEA shall set the current element state to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag false to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			

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Satisfies:	ELO-BF:FUR:2190
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SEA-KLL:SSS:REQ: 4573	Line Keylock Client - DISTURBED - input channel changes - transition to ACKNOWLEDGING										
Given the current element state is DISTURBED And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [OFF, ON] within DISTURBANCE_TIMER ms Then SEA shall set the current element state to ACKNOWLEDGING And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP <table><tr><th>SupervisionInState</th><th>AckButtonInState</th></tr><tr><td>DISTURBED</td><td>DISTURBED</td></tr><tr><td>DISTURBED</td><td>OFF</td></tr><tr><td>ON</td><td>DISTURBED</td></tr></table>				SupervisionInState	AckButtonInState	DISTURBED	DISTURBED	DISTURBED	OFF	ON	DISTURBED
SupervisionInState	AckButtonInState										
DISTURBED	DISTURBED										
DISTURBED	OFF										
ON	DISTURBED										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW ECMUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:	This behaviour is applicable while the DISTURBANCE_TIMER is running / not running.										
Satisfies:	ELO-BF:FUR:2191, ELO-BF:FUR:2268										

SEA-KLL:SSS:REQ: 4508	Line Keylock Client - DISTURBED - DISTURBANCE_TIMER expiry - transition to ACKNOWLEDGING
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Given the current element state is DISTURBED
And the current M1_Key_Supervision_IN input channel is OFF
And the current M2_Ack_Button_IN input channel is ON

When the current state of the element has been DISTURBED for DISTURBANCE_TIMER ms

Then SEA shall set the current element state to ACKNOWLEDGING
And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2191		

SEA-KLL:SSS:REQ: 4510 **Line Keylock Client - QT_PRESSED - QT_SYNC_TIMER expiry - transition to DISTURBED**

When the current state of the element has been QT_PRESSED for QT_SYNC_TIMER ms

Then SEA shall set the current element state to DISTURBED
And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2189, ELO-BF:FUR:2018		

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SEA-KLL:SSS:REQ:4512	Line Keylock Client - QT_PRESSED - input channel change - transition to ACKNOWLEDGING											
Given the current element state is QT_PRESSED And the current M1_Key_Supervision_IN input channel is <State> And the current M2_Ack_Button_IN input channel is <State> When changing the <InputChannel> input channel to <NewState> Then SEA shall set the current element state to ACKNOWLEDGING And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP <table><tr><th>State</th><th>InputChannel</th><th>NewState</th></tr><tr><td>ON</td><td>M1_Key_Supervision_IN</td><td>OFF</td></tr><tr><td>OFF</td><td>M2_Ack_Button_IN</td><td>ON</td></tr></table>				State	InputChannel	NewState	ON	M1_Key_Supervision_IN	OFF	OFF	M2_Ack_Button_IN	ON
State	InputChannel	NewState										
ON	M1_Key_Supervision_IN	OFF										
OFF	M2_Ack_Button_IN	ON										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW ECMUX									
Verification Method:	Test	Verification Step:	module test, subit									
Comments:												
Satisfies:	EL4-VIL-SSS-REQ-01521											

SEA-KLL:SSS:REQ: 4529	Line Keylock Client - QT_PRESSED - input channel change - transition to DISTURBED
Given the current element state is QT_PRESSED And the current M1_Key_Supervision_IN input channel is <State> And the current M2_Ack_Button_IN input channel is <State> When changing the <InputChannel> input channel to <NewState>	

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Then SEA shall set the current element state to DISTURBED
And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP

State	InputChannel	NewState
ON	M2_Ack_Button_IN	OFF
OFF	M1_Key_Supervision_IN	ON

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4574	Line Keylock Client - QT_PRESSED - input channel changes - transition to DISTURBED			
Given the current element state is QT_PRESSED When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [<States>] Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><th>States</th></tr><tr><td>OFF, OFF</td></tr><tr><td>ON, ON</td></tr></table>		States	OFF, OFF	ON, ON
States				
OFF, OFF				
ON, ON				
Type:	FUR - Functional Requirement			

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2189, ELO-BF:FUR:2018		

SEA-KLL:SSS:REQ: 4513	Line Keylock Client - ACKNOWLEDGING - ACK_STORE_TIMER expiry - transition to DISTURBED					
Given the current element state is ACKNOWLEDGING And the current M1_Key_Supervision_IN input channel is <State> And the current M2_Ack_Button_IN input channel is <State> When the current state of the element has been ACKNOWLEDGING for ACK_STORE_TIMER ms Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><td>State</td></tr><tr><td>ON</td></tr><tr><td>OFF</td></tr></table>				State	ON	OFF
State						
ON						
OFF						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:	ELO-BF:FUR:2320, ELO-BF:FUR:2189, ELO-BF:FUR:2018					

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SEA-KLL:SSS:REQ: 4514	Line Keylock Client - ACKNOWLEDGING - ACK_STORE_TIMER expiry - transition to DISTURBED		
Given the current element state is ACKNOWLEDGING And the current M1_Key_Supervision_IN input channel is OFF And the current M2_Ack_Button_IN input channel is ON When the current state of the element has been ACKNOWLEDGING for ACK_STORE_TIMER ms Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2320		

SEA-KLL:SSS:REQ: 4515	Line Keylock Client - ACKNOWLEDGING - input channel change - do nothing																		
Given the current element state is ACKNOWLEDGING And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to <State> Then SEA shall do nothing																			
<table><tr><th>SupervisionInState</th><th>AckButtonInState</th><th>InputChannel</th><th>State</th></tr><tr><td>OFF</td><td>ON</td><td>M1_ Key_ Supervision_ IN</td><td>ON</td></tr><tr><td>OFF</td><td>ON</td><td>M2_ Ack_ Button_ IN</td><td>OFF</td></tr><tr><td></td><td></td><td>M1_ Key_ Supervision_</td><td></td></tr></table>				SupervisionInState	AckButtonInState	InputChannel	State	OFF	ON	M1_ Key_ Supervision_ IN	ON	OFF	ON	M2_ Ack_ Button_ IN	OFF			M1_ Key_ Supervision_	
SupervisionInState	AckButtonInState	InputChannel	State																
OFF	ON	M1_ Key_ Supervision_ IN	ON																
OFF	ON	M2_ Ack_ Button_ IN	OFF																
		M1_ Key_ Supervision_																	

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SupervisionInState	AckButtonInState	InputChannel	State
ON	ON	IN	OFF
OFF	OFF	M2_Ack_Button_IN	ON

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4575	Line Keylock Client - ACKNOWLEDGING - input channel changes - do nothing								
Given the current element state is ACKNOWLEDGING And the current M1_Key_Supervision_IN input channel is <State> And the current M2_Ack_Button_IN input channel is <State> When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [<NewStates>] Then SEA shall do nothing <table><tr><th>State</th><th>NewStates</th></tr><tr><td>ON</td><td>OFF, OFF</td></tr><tr><td>OFF</td><td>ON, ON</td></tr></table>				State	NewStates	ON	OFF, OFF	OFF	ON, ON
State	NewStates								
ON	OFF, OFF								
OFF	ON, ON								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW ECMUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:									

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Satisfies:	EL4-VIL-SSS-REQ-01521
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SEA-KLL:SSS:REQ:4516	Line Keylock Client - ACKNOWLEDGING - input channel changes - transition to ACKNOWLEDGED														
Given the current element state is ACKNOWLEDGING And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to <State> Then SEA shall set the current element state to ACKNOWLEDGED															
<table><tr><th>SupervisionInState</th><th>AckButtonInState</th><th>InputChannel</th><th>State</th></tr><tr><td>ON</td><td>ON</td><td>M2_Ack_Button_IN</td><td>OFF</td></tr><tr><td>OFF</td><td>OFF</td><td>M1_Key_Supervision_IN</td><td>ON</td></tr></table>				SupervisionInState	AckButtonInState	InputChannel	State	ON	ON	M2_Ack_Button_IN	OFF	OFF	OFF	M1_Key_Supervision_IN	ON
SupervisionInState	AckButtonInState	InputChannel	State												
ON	ON	M2_Ack_Button_IN	OFF												
OFF	OFF	M1_Key_Supervision_IN	ON												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW ECMUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:															
Satisfies:	EL4-VIL-SSS-REQ-01521														

SEA-KLL:SSS:REQ: 4536	Line Keylock Client - ACKNOWLEDGING - input channel changes - transition to ACKNOWLEDGED		
Given the current element state is ACKNOWLEDGING When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [ON, OFF] Then SEA shall set the current element state to ACKNOWLEDGED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4517	Line Keylock Client - ACKNOWLEDGED - Locked command - transition to LOCKED		
Given the current element state is ACKNOWLEDGED When receiving the command e_key_locked_p from CC Then SEA shall set the current element state to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag true to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4537	Line Keylock Client - ACKNOWLEDGED - input channel changes - transition to ACKNOWLEDGING
Given the current element state is ACKNOWLEDGED When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [OFF, ON] Then SEA shall set the current element state to ACKNOWLEDGING	
Type:	FUR - Functional Requirement

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4518	Line Keylock Client - ACKNOWLEDGED - input channel change - transition to WITHDRAWING								
Given the current element state is ACKNOWLEDGED When changing the <InputChannel> input channel to <State> Then SEA shall set the current element state to WITHDRAWING <table><tr><th>InputChannel</th><th>State</th></tr><tr><td>M1_ Key_ Supervision_ IN</td><td>OFF</td></tr><tr><td>M2_ Ack_ Button_ IN</td><td>ON</td></tr></table>				InputChannel	State	M1_ Key_ Supervision_ IN	OFF	M2_ Ack_ Button_ IN	ON
InputChannel	State								
M1_ Key_ Supervision_ IN	OFF								
M2_ Ack_ Button_ IN	ON								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW ECMUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:									
Satisfies:	EL4-VIL-SSS-REQ-01521								

SEA-KLL:SSS:REQ: 4519	Line Keylock Client - WITHDRAWING - SYNC_TIMER expiry - transition to DISTURBED
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Given the current element state is WITHDRAWING
And the current M1_Key_Supervision_IN input channel is ON
And the current M2_Ack_Button_IN input channel is ON

When the current state of the element has been WITHDRAWING for SYNC_TIMER ms

Then SEA shall set the current element state to DISTURBED
And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2018		

SEA-KLL:SSS:REQ: 4520 **Line Keylock Client - WITHDRAWING - Locked command - transition to WAIT_FOR_LOCK**

Given the current element state is WITHDRAWING

When receiving the command e_key_locked_p from CC

Then SEA shall set the current element state to WAIT_FOR_LOCK
And SEA shall report the element state e_key_locked_es with command reply flag true to CC
And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			

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Satisfies:	EL4-VIL-SSS-REQ-01521
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SEA-KLL:SSS:REQ: 4521	Line Keylock Client - WITHDRAWING - SYNC_TIMER expiry - transition to WITHDRAWN		
Given the current element state is WITHDRAWING And the current M1_Key_Supervision_IN input channel is OFF And the current M2_Ack_Button_IN input channel is OFF When the current state of the element has been WITHDRAWING for SYNC_TIMER ms Then SEA shall set the current element state to WITHDRAWN And SEA shall report the element state e_key_withdrawn_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_WITHDRAWN to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4530	Line Keylock Client - WITHDRAWING - input channel change - transition to ACKNOWLEDGED														
Given the current element state is WITHDRAWING And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to <State> Then SEA shall set the current element state to ACKNOWLEDGED															
<table><tr><th>SupervisionInState</th><th>AckButtonInState</th><th>InputChannel</th><th>State</th></tr><tr><td>ON</td><td>ON</td><td>M2_Ack_Button_IN</td><td>OFF</td></tr><tr><td></td><td></td><td>M1_Key_Supervision_</td><td></td></tr></table>				SupervisionInState	AckButtonInState	InputChannel	State	ON	ON	M2_Ack_Button_IN	OFF			M1_Key_Supervision_	
SupervisionInState	AckButtonInState	InputChannel	State												
ON	ON	M2_Ack_Button_IN	OFF												
		M1_Key_Supervision_													

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SupervisionInState	AckButtonInState	InputChannel	State
OFF	OFF	IN	ON
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4531	Line Keylock Client - WITHDRAWING - input channel change - transition to ACKNOWLEDGING		
Given the current element state is WITHDRAWING And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to <State> Then SEA shall set the current element state to ACKNOWLEDGING			
SupervisionInState	AckButtonInState	InputChannel	State
ON	ON	M1_Key_Supervision_IN	OFF
OFF	OFF	M2_Ack_Button_IN	ON
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			

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Satisfies:	EL4-VIL-SSS-REQ-01521
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SEA-KLL:SSS:REQ: 4576	Line Keylock Client - WITHDRAWING - input channel changes - transition to DISTURBED					
Given the current element state is WITHDRAWING When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [<States>] Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><td>States</td></tr><tr><td>OFF, OFF</td></tr><tr><td>ON, ON</td></tr></table>				States	OFF, OFF	ON, ON
States						
OFF, OFF						
ON, ON						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:	ELO-BF:FUR:2189, ELO-BF:FUR:2018					

SEA-KLL:SSS:REQ: 4532	Line Keylock Client - WAIT_FOR_LOCK - SYNC_TIMER expiry - transition to DISTURBED
Given the current element state is WAIT_FOR_LOCK When the current state of the element has been WAIT_FOR_LOCK for SYNC_TIMER ms	

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Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag true to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2189, ELO-BF:FUR:2018		

SEA-KLL:SSS:REQ:4533	Line Keylock Client - WAIT_FOR_LOCK - input channel change - do nothing																						
Given the current element state is WAIT_FOR_LOCK And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to <State> Then SEA shall do nothing																							
<table><tr><th>SupervisionInState</th><th>AckButtonInState</th><th>InputChannel</th><th>State</th></tr><tr><td>ON</td><td>ON</td><td>M1_Key_Supervision_IN</td><td>OFF</td></tr><tr><td>OFF</td><td>OFF</td><td>M2_Ack_Button_IN</td><td>ON</td></tr><tr><td>OFF</td><td>ON</td><td>M1_Key_Supervision_IN</td><td>ON</td></tr><tr><td>OFF</td><td>ON</td><td>M2_Ack_Button_IN</td><td>OFF</td></tr></table>				SupervisionInState	AckButtonInState	InputChannel	State	ON	ON	M1_Key_Supervision_IN	OFF	OFF	OFF	M2_Ack_Button_IN	ON	OFF	ON	M1_Key_Supervision_IN	ON	OFF	ON	M2_Ack_Button_IN	OFF
SupervisionInState	AckButtonInState	InputChannel	State																				
ON	ON	M1_Key_Supervision_IN	OFF																				
OFF	OFF	M2_Ack_Button_IN	ON																				
OFF	ON	M1_Key_Supervision_IN	ON																				
OFF	ON	M2_Ack_Button_IN	OFF																				
Type:		FUR - Functional Requirement																					
Status:	draft	Stability:	stable																				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																				
Originator:	Technics (R&D)	Allocation:	SW ECMUX																				

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Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4579	Line Keylock Client - WAIT_FOR_LOCK - input channel changes - do nothing					
Given the current element state is WAIT_FOR_LOCK When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [<States>] Then SEA shall do nothing <table><tr><th>States</th></tr><tr><td>OFF, OFF</td></tr><tr><td>ON, ON</td></tr></table>				States	OFF, OFF	ON, ON
States						
OFF, OFF						
ON, ON						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:	EL4-VIL-SSS-REQ-01521					

SEA-KLL:SSS:REQ: 4534	Line Keylock Client - WAIT_FOR_LOCK - input channel change - transition to LOCKED
Given the current element state is WAIT_FOR_LOCK And the current M1_Key_Supervision_IN input channel is <SupervisionInState> And the current M2_Ack_Button_IN input channel is <AckButtonInState> When changing the <InputChannel> input channel to <State>	

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Then SEA shall set the current element state to LOCKED			
SupervisionInState	AckButtonInState	InputChannel	State
ON	ON	M2_ Ack_ Button_ IN	OFF
OFF	OFF	M1_ Key_ Supervision_ IN	ON
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4580	Line Keylock Client - WAIT_FOR_LOCK - input channel changes - transition to LOCKED		
Given the current element state is WAIT_FOR_LOCK			
When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [ON, OFF]			
Then SEA shall set the current element state to LOCKED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ:	Line Keylock Client - WITHDRAWN - input channel changes - transition
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4577	to DISTURBED		
Given the current element state is WITHDRAWN When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [ON, ON] Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2018		

SEA-KLL:SSS:REQ: 4522	Line Keylock Client - WITHDRAWN - input channel change - transition to LOCKING					
Given the current element state is WITHDRAWN When changing the <InputChannel> input channel to ON Then SEA shall set the current element state to LOCKING <table><tr><th>InputChannel</th></tr><tr><td>M1_ Key_ Supervision_ IN</td></tr><tr><td>M2_ Ack_ Button_ IN</td></tr></table>				InputChannel	M1_ Key_ Supervision_ IN	M2_ Ack_ Button_ IN
InputChannel						
M1_ Key_ Supervision_ IN						
M2_ Ack_ Button_ IN						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			
Verification Method:	Test	Verification Step:	module test, subit			

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Comments:	
Satisfies:	EL4-VIL-SSS-REQ-01521

SEA-KLL:SSS:REQ: 4523	Line Keylock Client - LOCKING - SYNC_TIMER expiry - transition to LOCKED		
Given the current element state is LOCKING And the current M1_Key_Supervision_IN input channel is ON And the current M2_Ack_Button_IN input channel is OFF When the current state of the element has been LOCKING for SYNC_TIMER ms Then SEA shall set the current element state to LOCKED And SEA shall report the element state e_key_locked_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4524	Line Keylock Client - LOCKING - SYNC_TIMER expiry - transition to ACKNOWLEDGING
Given the current element state is LOCKING And the current M1_Key_Supervision_IN input channel is OFF And the current M2_Ack_Button_IN input channel is ON When the current state of the element has been LOCKING for SYNC_TIMER ms Then SEA shall set the current element state to ACKNOWLEDGING And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4525	Line Keylock Client - LOCKING - input channel change - transition to WITHDRAWN					
Given the current element state is LOCKING When changing the <InputChannel> input channel to OFF Then SEA shall set the current element state to WITHDRAWN <table><tr><th>InputChannel</th></tr><tr><td>M1_ Key_ Supervision_ IN</td></tr><tr><td>M2_ Ack_ Button_ IN</td></tr></table>				InputChannel	M1_ Key_ Supervision_ IN	M2_ Ack_ Button_ IN
InputChannel						
M1_ Key_ Supervision_ IN						
M2_ Ack_ Button_ IN						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:	EL4-VIL-SSS-REQ-01521					

SEA-KLL:SSS:REQ: 4527	Line Keylock Client - LOCKING - input channel change - transition to DISTURBED
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Given the current element state is LOCKING

When changing the <InputChannel> input channel to ON

Then SEA shall set the current element state to DISTURBED

And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC

And SEA shall send the fault notification sum_fault with fault type e_if_error to CC

And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP

InputChannel

M1_Key_Supervision_
IN

M2_Ack_Button_IN

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2018		

SEA-KLL:SSS:REQ: 4578	Line Keylock Client - LOCKING - input channel changes - transition to DISTURBED
----------------------------------	--

Given the current element state is LOCKING

When changing the input channels [M1_Key_Supervision_IN, M2_Ack_Button_IN] to [<States>]

Then SEA shall set the current element state to DISTURBED

And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC

And SEA shall send the fault notification sum_fault with fault type e_if_error to CC

And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP

States

OFF,
ON

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States			
	ON, OFF		
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4526	Line Keylock Client - commands - do nothing																				
Given the current element state is <State> When receiving the command <Command> from CC Then SEA shall do nothing <table><tr><th>State</th><th>Command</th></tr><tr><td>DISTURBED</td><td>e_key_locked_p</td></tr><tr><td>DISTURBED</td><td>e_key_acknowledged_p</td></tr><tr><td>LOCKED</td><td>e_key_locked_p</td></tr><tr><td>QT_PRESSED</td><td>e_key_acknowledged_p</td></tr><tr><td>QT_PRESSED</td><td>e_key_locked_p</td></tr><tr><td>ACKNOWLEDGING</td><td>e_key_locked_p</td></tr><tr><td>ACKNOWLEDGING</td><td>e_key_acknowledged_p</td></tr><tr><td>ACKNOWLEDGED</td><td>e_key_acknowledged_p</td></tr><tr><td></td><td>e_key_acknowledged_p</td></tr></table>		State	Command	DISTURBED	e_key_locked_p	DISTURBED	e_key_acknowledged_p	LOCKED	e_key_locked_p	QT_PRESSED	e_key_acknowledged_p	QT_PRESSED	e_key_locked_p	ACKNOWLEDGING	e_key_locked_p	ACKNOWLEDGING	e_key_acknowledged_p	ACKNOWLEDGED	e_key_acknowledged_p		e_key_acknowledged_p
State	Command																				
DISTURBED	e_key_locked_p																				
DISTURBED	e_key_acknowledged_p																				
LOCKED	e_key_locked_p																				
QT_PRESSED	e_key_acknowledged_p																				
QT_PRESSED	e_key_locked_p																				
ACKNOWLEDGING	e_key_locked_p																				
ACKNOWLEDGING	e_key_acknowledged_p																				
ACKNOWLEDGED	e_key_acknowledged_p																				
	e_key_acknowledged_p																				

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State	Command
WITHDRAWING	p
WAIT_FOR_LOCK	e_key_locked_p
WAIT_FOR_LOCK	e_key_acknowledged_p
WITHDRAWN	e_key_locked_p
WITHDRAWN	e_key_acknowledged_p
LOCKING	e_key_locked_p
LOCKING	e_key_acknowledged_p

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ:4528	Line Keylock Client - Safety reaction when e_key_prerel_norm_p / e_key_prerel_canc_p / e_key_rel_norm_p / e_key_withdrawn_p command is received				
When receiving the command <Command> from CC Then SEA shall execute ECMUX safety reaction Deactivation <table><tr><th>Command</th></tr><tr><td>e_key_prerel_norm_p</td></tr><tr><td>e_key_prerel_canc_p</td></tr><tr><td>e_key_rel_norm_p</td></tr></table>		Command	e_key_prerel_norm_p	e_key_prerel_canc_p	e_key_rel_norm_p
Command					
e_key_prerel_norm_p					
e_key_prerel_canc_p					
e_key_rel_norm_p					

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<table><tr><th>Command</th></tr><tr><td>e_key_withdrawn_p</td></tr></table>				Command	e_key_withdrawn_p
Command					
e_key_withdrawn_p					
Type:	FUR - Functional Requirement				
Status:	draft	Stability:	stable		
Target Release:	SEA SW GEN 3.0	Priority:	mandatory		
Originator:	Technics (R&D)	Allocation:	SW ECMUX		
Verification Method:	Test	Verification Step:	module test, subit		
Comments:					
Satisfies:	EL4-VIL-SSS-REQ-01521				

SEA-KLL:SSS:REQ: 4511	Line Keylock Client - unexpected input channel changes - transition to DISTURBED																
Given the current element state is <State> When changing the <InputChannel> input channel to DISTURBED Then SEA shall set the current element state to DISTURBED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP <table><tr><th>State</th><th>InputChannel</th></tr><tr><td>LOCKED</td><td>M1_Key_Supervision_IN</td></tr><tr><td>LOCKED</td><td>M2_Ack_Button_IN</td></tr><tr><td>QT_PRESSED</td><td>M1_Key_Supervision_IN</td></tr><tr><td>QT_PRESSED</td><td>M2_Ack_Button_IN</td></tr><tr><td>ACKNOWLEDGING</td><td>M1_Key_Supervision_IN</td></tr><tr><td>ACKNOWLEDGING</td><td>M2_Ack_Button_IN</td></tr><tr><td>ACKNOWLEDGED</td><td>M1_Key_Supervision_IN</td></tr></table>		State	InputChannel	LOCKED	M1_Key_Supervision_IN	LOCKED	M2_Ack_Button_IN	QT_PRESSED	M1_Key_Supervision_IN	QT_PRESSED	M2_Ack_Button_IN	ACKNOWLEDGING	M1_Key_Supervision_IN	ACKNOWLEDGING	M2_Ack_Button_IN	ACKNOWLEDGED	M1_Key_Supervision_IN
State	InputChannel																
LOCKED	M1_Key_Supervision_IN																
LOCKED	M2_Ack_Button_IN																
QT_PRESSED	M1_Key_Supervision_IN																
QT_PRESSED	M2_Ack_Button_IN																
ACKNOWLEDGING	M1_Key_Supervision_IN																
ACKNOWLEDGING	M2_Ack_Button_IN																
ACKNOWLEDGED	M1_Key_Supervision_IN																

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State	InputChannel
ACKNOWLEDGED	M2_Ack_Button_IN
WITHDRAWING	M1_Key_Supervision_IN
WITHDRAWING	M2_Ack_Button_IN
WAIT_FOR_LOCK	M1_Key_Supervision_IN
WAIT_FOR_LOCK	M2_Ack_Button_IN
WITHDRAWN	M1_Key_Supervision_IN
WITHDRAWN	M2_Ack_Button_IN
LOCKING	M1_Key_Supervision_IN
LOCKING	M2_Ack_Button_IN

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

3.1.1.2.2 Connection - Line Keylock - Client

In order to control and supervise a Line Keylock Client

As a SEA

I want to be able to supervise connections and when changes occur then

- to execute commands during update_history phase, attempt to repair the commanded state and report it to CC
- on answering the general inquiry request from CC report the last stable element state to CC
- on CC connection establishment report the last stable element state to CC
- on OC-IO connection loss take on the default state DISCONNECTED
- on OC-IO connection establishment transition to LOCKED / ACKNOWLEDGING / WITHDRAWN /

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DISTURBED state depending on the input channels state
- report the connection state changes to CC and DGP

For requirement language see Chapter "Requirement Language".

Given SEA with a configured Line Keylock Client element

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
KEY_LOCKED	9
KEY_DISTURBED	11
KEY_ACKNOWLEDGED	29
KEY_WITHDRAWN	10

SEA-KLL:SSS:REQ: 4540	Line Keylock Client - Ignore commands e_key_locked_p / e_key_acknowledged_p during update_history phase																		
Given the current element state is <State> And CC connection state is UPDATE_HISTORY When receiving the command <Command> from CC Then SEA shall do nothing <table> <tr> <th>State</th><th>Command</th></tr> <tr> <td>DISCONNECTED</td><td>e_key_locked_p</td></tr> <tr> <td>DISCONNECTED</td><td>e_key_acknowledged_p</td></tr> <tr> <td>LOCKED</td><td>e_key_locked_p</td></tr> <tr> <td>QT_PRESSED</td><td>e_key_locked_p</td></tr> <tr> <td>QT_PRESSED</td><td>e_key_acknowledged_p</td></tr> <tr> <td>ACKNOWLEDGING</td><td>e_key_locked_p</td></tr> <tr> <td>ACKNOWLEDGING</td><td>e_key_acknowledged_p</td></tr> <tr> <td>ACKNOWLEDGED</td><td>e_key_acknowledged_p</td></tr> </table>		State	Command	DISCONNECTED	e_key_locked_p	DISCONNECTED	e_key_acknowledged_p	LOCKED	e_key_locked_p	QT_PRESSED	e_key_locked_p	QT_PRESSED	e_key_acknowledged_p	ACKNOWLEDGING	e_key_locked_p	ACKNOWLEDGING	e_key_acknowledged_p	ACKNOWLEDGED	e_key_acknowledged_p
State	Command																		
DISCONNECTED	e_key_locked_p																		
DISCONNECTED	e_key_acknowledged_p																		
LOCKED	e_key_locked_p																		
QT_PRESSED	e_key_locked_p																		
QT_PRESSED	e_key_acknowledged_p																		
ACKNOWLEDGING	e_key_locked_p																		
ACKNOWLEDGING	e_key_acknowledged_p																		
ACKNOWLEDGED	e_key_acknowledged_p																		

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State	Command
WITHDRAWING	e_key_locked_p
WITHDRAWING	e_key_acknowledged_p
WAIT_FOR_LOCK	e_key_locked_p
WAIT_FOR_LOCK	e_key_acknowledged_p
WITHDRAWN	e_key_locked_p
WITHDRAWN	e_key_acknowledged_p
LOCKING	e_key_locked_p
LOCKING	e_key_acknowledged_p

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4541	Line Keylock Client - Execute command e_key_acknowledged_p during update_history phase in LOCKED - transition to ACKNOWLEDGED		
Given the current element state is LOCKED And CC connection state is UPDATE_HISTORY When receiving the command e_key_acknowledged_p from CC Then SEA shall set the current element state to ACKNOWLEDGED And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4542	Line Keylock Client - Execute command e_key_locked_p during update_history phase in ACKNOWLEDGED - transition to LOCKED		
Given the current element state is ACKNOWLEDGED And CC connection state is UPDATE_HISTORY When receiving the command e_key_locked_p from CC Then SEA shall set the current element state to LOCKED And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4550	Line Keylock Client - General Inquiry request from CC
Given the current element state is <State> And CC connection state is GA_STARTED When answering the general inquiry request from CC Then SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall send the fault notification <SumFault> with fault type e_if_error to CC	

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State	CCState	SumFault
DISCONNECTED	e_key_disturbed_es	sum_fault
DISTURBED	e_key_disturbed_es	sum_fault
LOCKED	e_key_locked_es	sum_fault_rep
QT_PRESSED	e_key_locked_es	sum_fault_rep
WAIT_FOR_LOCK	e_key_locked_es	sum_fault_rep
ACKNOWLEDGING	e_key_acknowledged_es	sum_fault_rep
ACKNOWLEDGED	e_key_acknowledged_es	sum_fault_rep
WITHDRAWING	e_key_acknowledged_es	sum_fault_rep
WITHDRAWN	e_key_withdrawn_es	sum_fault_rep
LOCKING	e_key_withdrawn_es	sum_fault_rep

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Note that before answering the general inquiry request, other stages of CC connection establishment are completed. Between connection establishment start and the general inquiry, element state maybe be updated if there is a command from CC received during update_history phase.		
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ:4551	Line Keylock Client - CC connection loss - do nothing
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Given the current element state is <State>
And CC connection state is CONNECTED

When losing the connection to CC

Then SEA shall do nothing

State
DISCONNECTED
DISTURBED
LOCKED
QT_ PRESSED
ACKNOWLEDGING
ACKNOWLEDGED
WITHDRAWING
WAIT_ FOR_ LOCK
WITHDRAWN
LOCKING

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4552	Line Keylock Client - [Re-]Connecting to OC-IO in DISCONNECTED - transition to LOCKED
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Given the current element state is DISCONNECTED
And ECMUX - OC-IO connection state is NOT_CONNECTED
And CC connection state is CONNECTED
And the OC-IO state of the input channel M1_Key_Supervision_IN evaluates to ON
And the OC-IO state of the input channel M2_Ack_Button_IN evaluates to OFF

When establishing the connection between ECMUX and OC-IO

Then SEA shall set the current element state to LOCKED
And SEA shall report the element state e_key_locked_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_LOCKED to DGP
And SEA shall report the connection_ok diagnosis device fault repair message to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2015		

SEA-KLL:SSS:REQ: 4553 **Line Keylock Client - [Re-]Connecting to OC-IO in DISCONNECTED - transition to ACKNOWLEDGING**

Given the current element state is DISCONNECTED
And ECMUX - OC-IO connection state is NOT_CONNECTED
And CC connection state is CONNECTED
And the OC-IO state of the input channel M1_Key_Supervision_IN evaluates to OFF
And the OC-IO state of the input channel M2_Ack_Button_IN evaluates to ON

When establishing the connection between ECMUX and OC-IO

Then SEA shall set the current element state to ACKNOWLEDGING
And SEA shall report the element state e_key_acknowledged_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message KEY_ACKNOWLEDGED to DGP
And SEA shall report the connection_ok diagnosis device fault repair message to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2016		

SEA-KLL:SSS:REQ: 4554	Line Keylock Client - [Re-]Connecting to OC-IO in DISCONNECTED - transition to WITHDRAWN		
Given the current element state is DISCONNECTED And ECMUX - OC-IO connection state is NOT_CONNECTED And CC connection state is CONNECTED And the OC-IO state of the input channel M1_Key_Supervision_IN evaluates to OFF And the OC-IO state of the input channel M2_Ack_Button_IN evaluates to OFF When establishing the connection between ECMUX and OC-IO Then SEA shall set the current element state to WITHDRAWN And SEA shall report the element state e_key_withdrawn_es with command reply flag false to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_WITHDRAWN to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2017		

SEA-KLL:SSS:REQ: 4555	Line Keylock Client - [Re-]Connecting to OC-IO in DISCONNECTED - transition to DISTURBED		
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Given the current element state is DISCONNECTED
And ECMUX - OC-IO connection state is NOT_CONNECTED
And CC connection state is CONNECTED
And the OC-IO state of the input channel M1_Key_Supervision_IN evaluates to <SupervisionInState>
And the OC-IO state of the input channel M2_Ack_Button_IN evaluates to <AckButtonInState>

When establishing the connection between ECMUX and OC-IO

Then SEA shall set the current element state to DISTURBED
And SEA shall report the connection_ok diagnosis device fault repair message to DGP

SupervisionInState	AckButtonInState
ON	ON
ON	DISTURBED
OFF	DISTURBED
DISTURBED	ON
DISTURBED	OFF
DISTURBED	DISTURBED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	ELO-BF:FUR:2065		

SEA-KLL:SSS:REQ: 4556	Line Keylock Client - Connection loss to OC-IO - DISTURBED - transition to DISCONNECTED
Given the current element state is DISTURBED And ECMUX - OC-IO connection state is CONNECTED When losing the connection between ECMUX and OC-IO Then SEA shall set the current element state to DISCONNECTED And SEA shall report the no_connection diagnosis device fault message to DGP	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4557	Line Keylock Client - Connection loss to OC-IO - transition to DISCONNECTED											
Given the current element state is <State> And ECMUX - OC-IO connection state is CONNECTED When losing the connection between ECMUX and OC-IO Then SEA shall set the current element state to DISCONNECTED And SEA shall report the element state e_key_disturbed_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><th>State</th></tr><tr><td>LOCKED</td></tr><tr><td>QT_PRESSED</td></tr><tr><td>ACKNOWLEDGING</td></tr><tr><td>ACKNOWLEDGED</td></tr><tr><td>WITHDRAWING</td></tr><tr><td>WAIT_FOR_LOCK</td></tr><tr><td>WITHDRAWN</td></tr><tr><td>LOCKING</td></tr></table>				State	LOCKED	QT_PRESSED	ACKNOWLEDGING	ACKNOWLEDGED	WITHDRAWING	WAIT_FOR_LOCK	WITHDRAWN	LOCKING
State												
LOCKED												
QT_PRESSED												
ACKNOWLEDGING												
ACKNOWLEDGED												
WITHDRAWING												
WAIT_FOR_LOCK												
WITHDRAWN												
LOCKING												
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:	EL4-VIL-SSS-REQ-01521		

SEA-KLL:SSS:REQ: 4558	Line Keylock Client - Ignore command e_key_locked_p / e_key_acknowledged_p in DISCONNECTED - do nothing					
Given the current element state is DISCONNECTED And ECMUX - OC-IO connection state is NOT_CONNECTED When receiving the command <Command> from CC Then SEA shall do nothing <table><tr><th>Command</th></tr><tr><td>e_key_locked_p</td></tr><tr><td>e_key_acknowledged_p</td></tr></table>				Command	e_key_locked_p	e_key_acknowledged_p
Command						
e_key_locked_p						
e_key_acknowledged_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW ECMUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:	EL4-VIL-SSS-REQ-01521					

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3.1.1.2.3 Diagnosis - Line Keylock - Client

In order to provide the aspect diagnosis for a Line Keylock Client
As a SEA
I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried element

For requirement language see Chapter "Requirement Language".

Given SEA with a configured Line Keylock Client element
And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
KEY_ LOCKED	9
KEY_ DISTURBED	11
KEY_ ACKNOWLEDGED	29
KEY_ WITHDRAWN	10

SEA-KLL:SSS:REQ: 4560	Line Keylock Client - aspect query - OC-IO CONNECTED														
Given the current element state is <State> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>DGPMMessage</th></tr><tr><td>DISTURBED</td><td>KEY_ DISTURBED</td></tr><tr><td>LOCKED</td><td>KEY_ LOCKED</td></tr><tr><td>QT_ PRESSED</td><td>KEY_ LOCKED</td></tr><tr><td>WAIT_ FOR_ LOCK</td><td>KEY_ LOCKED</td></tr><tr><td>ACKNOWLEDGING</td><td>KEY_ ACKNOWLEDGED</td></tr><tr><td>ACKNOWLEDGED</td><td>KEY_ ACKNOWLEDGED</td></tr></table>		State	DGPMMessage	DISTURBED	KEY_ DISTURBED	LOCKED	KEY_ LOCKED	QT_ PRESSED	KEY_ LOCKED	WAIT_ FOR_ LOCK	KEY_ LOCKED	ACKNOWLEDGING	KEY_ ACKNOWLEDGED	ACKNOWLEDGED	KEY_ ACKNOWLEDGED
State	DGPMMessage														
DISTURBED	KEY_ DISTURBED														
LOCKED	KEY_ LOCKED														
QT_ PRESSED	KEY_ LOCKED														
WAIT_ FOR_ LOCK	KEY_ LOCKED														
ACKNOWLEDGING	KEY_ ACKNOWLEDGED														
ACKNOWLEDGED	KEY_ ACKNOWLEDGED														

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State	DGPMessage		
WITHDRAWING	KEY_ ACKNOWLEDGED		
WITHDRAWN	KEY_ WITHDRAWN		
LOCKING	KEY_ WITHDRAWN		
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

SEA-KLL:SSS:REQ: 4561	Line Keylock Client - aspect query - DISCONNECTED - OC-IO NOT_CONNECTED		
Given the current element state is DISCONNECTED			
When receiving an aspect query from DGP			
Then SEA shall report the diagnosis aspect message KEY_DISTURBED to DGP			
And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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3.2 Technical Requirements

3.2.1 Syntax checks

Incoming message syntax checks are defined generically for all Eulynx elements in SSS SEA GEN [5] Chapter 3.3.4.1 “Incoming Message Syntax checks”.

3.2.2 State Transfer - Line Keylock Server

In order to provide the internal state of a Line Keylock Server
As a SEA
I want to be able to serialize/deserialize element state when a state transfer during ECMUX redundancy integration takes place.

Given SEA with a configured Line Keylock Server element

SEA-KLL:SSS:REQ: 4460	Line Keylock Server - state transfer		
When a state transfer during ECMUX redundancy integration takes place			
Then the element process shall serialize/deserialize element state considering: internal element state			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test, Inspection/Review	Verification Step:	module test, code reading
Comments:	For the Line Keylock Server, during an ECMUX redundancy integration when the state transfer takes place the element process has to be capable to serialize a state and deserialize from a state with the information of the current internal element state.		
Satisfies:			

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3.2.3 System - Line Keylock - Client

In order to provide the internal state of a Line Keylock Client
As a SEA
I want to be able to serialize/deserialize element state when a state transfer during ECMUX redundancy integration takes place.

Given SEA with a configured Line Keylock Client element

SEA-KLL:SSS:REQ: 4570	Line Keylock Client - state transfer		
When a state transfer during ECMUX redundancy integration takes place			
Then the element process shall serialize/deserialize element state considering: internal element state			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW ECMUX
Verification Method:	Test, Inspection/Review	Verification Step:	module test, code reading
Comments:	For the Line Keylock Client, during an ECMUX redundancy integration when the state transfer takes place the element process has to be capable to serialize a state and deserialize from a state with the information of the current internal element state.		
Satisfies:			

3.2.4 Eulynx - SCI-IO

ECMUX-KLL:SSS:REQ: 4572	Keylocks - Implement SCI-IO
ECMUX shall implement the Keylocks based on Eulynx Baseline 4.4 as specified in the following documents:	

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- 20250619 Interface specification SCI-IO Eu.Doc.46 4.0 (4.A)			
- 20250619 Requirements specification for subsystem Generic IO Eu.Doc.45 v4.3 (1.A)			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Analysis	Verification Step:	Review
Comments:			
Satisfies:	EL4-DIL-SSS-REQ-00614		

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4 Requirements List

Identifier	Synopsis	Category	Status
SEA-KLL:SSS:REQ:4450	Line Keylock Server - aspect query - OC-IO CONNECTED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4452	Line Keylock Server - aspect query - DISCONNECTED - OC-IO NOT_CONNECTED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4500	Line Keylock Client - Initial state	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4501	Line Keylock Client - LOCKED - Acknowledged command - transition to ACKNOWLEDGED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4502	Line Keylock Client - LOCKED - input channel changes - transition to QT_PRESSED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4535	Line Keylock Client - LOCKED - input channel changes - transition to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4504	Line Keylock Client - DISTURBED - input channel changes to ON/OFF - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4505	Line Keylock Client - DISTURBED - input channel changes to DISTURBED - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4503	Line Keylock Client - DISTURBED - DISTURBANCE_TIMER expiry - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4506	Line Keylock Client - DISTURBED - input channel change - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4509	Line Keylock Client - DISTURBED - input channel change - transition to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4539	Line Keylock Client - DISTURBED - input channel changes - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4507	Line Keylock Client - DISTURBED - DISTURBANCE_TIMER expiry - transition to LOCKED	FUR - Functional Requirement	draft
		FUR - Functional	

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Identifier	Synopsis	Category	Status
SEA-KLL:SSS:REQ: 4573	Line Keylock Client - DISTURBED - input channel changes - transition to ACKNOWLEDGING	Requirement	draft
SEA-KLL:SSS:REQ: 4508	Line Keylock Client - DISTURBED - DISTURBANCE_TIMER expiry - transition to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4510	Line Keylock Client - QT_PRESSED - QT_SYNC_TIMER expiry - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4512	Line Keylock Client - QT_PRESSED - input channel change - transition to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4529	Line Keylock Client - QT_PRESSED - input channel change - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4574	Line Keylock Client - QT_PRESSED - input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4513	Line Keylock Client - ACKNOWLEDGING - ACK_STORE_TIMER expiry - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4514	Line Keylock Client - ACKNOWLEDGING - ACK_STORE_TIMER expiry - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4515	Line Keylock Client - ACKNOWLEDGING - input channel change - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4575	Line Keylock Client - ACKNOWLEDGING - input channel changes - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4516	Line Keylock Client - ACKNOWLEDGING - input channel changes - transition to ACKNOWLEDGED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4536	Line Keylock Client - ACKNOWLEDGING - input channel changes - transition to ACKNOWLEDGED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4517	Line Keylock Client - ACKNOWLEDGED - Locked command - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4537	Line Keylock Client - ACKNOWLEDGED - input channel changes - transition to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4518	Line Keylock Client - ACKNOWLEDGED - input channel change - transition to WITHDRAWING	FUR - Functional Requirement	draft
		FUR -	

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Identifier	Synopsis	Category	Status
SEA-KLL:SSS:REQ: 4519	Line Keylock Client - WITHDRAWING - SYNC_TIMER expiry - transition to DISTURBED	Functional Requirement	draft
SEA-KLL:SSS:REQ: 4520	Line Keylock Client - WITHDRAWING - Locked command - transition to WAIT_FOR_LOCK	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4521	Line Keylock Client - WITHDRAWING - SYNC_TIMER expiry - transition to WITHDRAWN	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4530	Line Keylock Client - WITHDRAWING - input channel change - transition to ACKNOWLEDGED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4531	Line Keylock Client - WITHDRAWING - input channel change - transition to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4576	Line Keylock Client - WITHDRAWING - input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4532	Line Keylock Client - WAIT_FOR_LOCK - SYNC_TIMER expiry - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4533	Line Keylock Client - WAIT_FOR_LOCK - input channel change - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4579	Line Keylock Client - WAIT_FOR_LOCK - input channel changes - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4534	Line Keylock Client - WAIT_FOR_LOCK - input channel change - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4580	Line Keylock Client - WAIT_FOR_LOCK - input channel changes - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4577	Line Keylock Client - WITHDRAWN - input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4522	Line Keylock Client - WITHDRAWN - input channel change - transition to LOCKING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4523	Line Keylock Client - LOCKING - SYNC_TIMER expiry - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4524	Line Keylock Client - LOCKING - SYNC_TIMER expiry - transition to ACKNOWLEDGING	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
SEA-KLL:SSS:REQ:4525	Line Keylock Client - LOCKING - input channel change - transition to WITHDRAWN	Functional Requirement	draft
SEA-KLL:SSS:REQ:4527	Line Keylock Client - LOCKING - input channel change - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4578	Line Keylock Client - LOCKING - input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4526	Line Keylock Client - commands - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4528	Line Keylock Client - Safety reaction when e_key_prerel_norm_p / e_key_prerel_canc_p / e_key_rel_norm_p / e_key_withdrawn_p command is received	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4511	Line Keylock Client - unexpected input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
ECMUX-KLL:SSS:REQ:4572	Keylocks - Implement SCI-IO	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4470	Line Keylock Server - Execute command e_key_prerel_norm_p during update_history phase in LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4471	Line Keylock Server - Execute command e_key_acknowledged_p during update_history phase in LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4472	Line Keylock Server - Execute command e_key_rel_norm_p during update_history phase in LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4473	Line Keylock Server - Ignore commands during update_history phase	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4401	Line Keylock Server - General Inquiry request from CC	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4402	Line Keylock Server - CC connection loss - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4403	Line Keylock Server - CC connection loss in PRE_RELEASED / RELEASED - switch to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ:4409	Line Keylock Server - CC connection loss in WITHDRAWN - switch to LOCKING	FUR - Functional Requirement	draft
		FUR -	

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Identifier	Synopsis	Category	Status
SEA-KLL:SSS:REQ: 4404	Line Keylock Server - DISCONNECTED - [Re-]Connecting to OC-IO - transition to LOCKED	Functional Requirement	draft
SEA-KLL:SSS:REQ: 4410	Line Keylock Server - DISCONNECTED - [Re-]Connecting to OC-IO - transition to LOCKING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4405	Line Keylock Server - DISCONNECTED - [Re-]Connecting to OC-IO - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4406	Line Keylock Server - Connection loss to OC-IO - DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4407	Line Keylock Server - Connection loss to OC-IO - transition to DISCONNECTED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4408	Line Keylock Server - DISCONNECTED - reject commands - OC-IO NOT_CONNECTED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4108	Station Keylock - Single Lock - Execute command e_key_rel_norm_p during update_history phase in LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4109	Station Keylock - Single Lock - Ignore commands during update_history phase	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4101	Station Keylock - Single Lock - DISCONNECTED / DISTURBED / WITHDRAWN_DISTURBED - General Inquiry request from CC - with fault	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4102	Station Keylock - Single Lock - LOCKING from DISCONNECTED / DISTURBED - General Inquiry request from CC - with fault	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4103	Station Keylock - Single Lock - LOCKING from LOCKED / RELEASED / WITHDRAWN - General Inquiry request from CC	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4105	Station Keylock - Single Lock - [Re-]Connecting to CC - without fault	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4106	Station Keylock - Single Lock - PREPARE_LOCKED - [Re-] Connecting to CC	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4110	Station Keylock - Single Lock - Connection loss to CC - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4114	Station Keylock - Single Lock - Connection loss to CC - switch to PREPARE_LOCKED	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
SEA-KLS:SSS:REQ: 4120	Station Keylock - Single Lock - DISCONNECTED - [Re-] Connecting to OC-IO - switch to LOCKING	Functional Requirement	draft
SEA-KLS:SSS:REQ: 4126	Station Keylock - Single Lock - DISCONNECTED - [Re-] Connecting to OC-IO - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4121	Station Keylock - Single Lock - LOCKING from DISCONNECTED / DISTURBED - Connection loss to OC-IO	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4122	Station Keylock - Single Lock - LOCKING from LOCKED / RELEASED / WITHDRAWN - Connection loss to OC-IO	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4104	Station Keylock - Single Lock - LOCKING from RELEASED - open command - Connection loss to OC-IO	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4124	Station Keylock - Single Lock - Connection loss to OC-IO - without fault reporting	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4125	Station Keylock - Single Lock - Connection loss to OC-IO - with fault reporting	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4107	Station Keylock - Single Lock - PREPARE_LOCKED from RELEASED - open command - Connection loss to OC-IO	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4119	Station Keylock - Single Lock - DISCONNECTED - Release / Cancel command - reject - OC-IO NOT_CONNECTED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4000	Station Keylock - Single Lock - Initial state	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4004	Station Keylock - Single Lock - LOCKING - Cancel command - reject	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4056	Station Keylock - Single Lock - LOCKING - Release command - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4005	Station Keylock - Single Lock - LOCKING - input channel change - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4006	Station Keylock - Single Lock - LOCKING from DISCONNECTED / DISTURBED - input channel change - switch to LOCKED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4067	Station Keylock - Single Lock - LOCKING from DISCONNECTED / DISTURBED - input channels change - switch to LOCKED	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
SEA-KLS:SSS:REQ: 4063	Station Keylock - Single Lock - LOCKING from LOCKED - input channel change - transition to LOCKED	Functional Requirement	draft
SEA-KLS:SSS:REQ: 4068	Station Keylock - Single Lock - LOCKING from LOCKED - input channels change - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4014	Station Keylock - Single Lock - LOCKING from RELEASED / WITHDRAWN - input channel change - switch to LOCKED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4069	Station Keylock - Single Lock - LOCKING from RELEASED / WITHDRAWN - input channels change - switch to LOCKED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4064	Station Keylock - Single Lock - LOCKING from RELEASED - input channel change - switch to LOCKED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4070	Station Keylock - Single Lock - LOCKING from RELEASED with open command e_key_rel_canc_p - input channels change - switch to LOCKED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4007	Station Keylock - Single Lock - LOCKING from DISCONNECTED / DISTURBED - input channel change to DISTURBED - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4008	Station Keylock - Single Lock - LOCKING from LOCKED / RELEASED / WITHDRAWN - input channel change to DISTURBED - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4009	Station Keylock - Single Lock - LOCKING from DISCONNECTED / DISTURBED - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4010	Station Keylock - Single Lock - LOCKING from LOCKED / RELEASED / WITHDRAWN - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4065	Station Keylock - Single Lock - LOCKING from RELEASED - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4011	Station Keylock - Single Lock - DISTURBED - Release / Cancel command - reject	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4012	Station Keylock - Single Lock - DISTURBED - OUT_CMD_DELAY_TIMER expiry - switch to LOCKING	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4013	Station Keylock - Single Lock - DISTURBED - OUT_CMD_DELAY_TIMER expiry - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4015	Station Keylock - Single Lock - DISTURBED - input channel changes - do nothing	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
SEA-KLS:SSS:REQ: 4016	Station Keylock - Single Lock - DISTURBED - input channel changes - switch to LOCKING	Functional Requirement	draft
SEA-KLS:SSS:REQ: 4017	Station Keylock - Single Lock - LOCKED - Cancel command - reject	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4018	Station Keylock - Single Lock - LOCKED - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4019	Station Keylock - Single Lock - LOCKED - Release command - switch to PREPARE_TEST	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4020	Station Keylock - Single Lock - PREPARE_TEST - Release / Cancel command - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4021	Station Keylock - Single Lock - PREPARE_TEST - RELEASE_TIMER expiry - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4022	Station Keylock - Single Lock - PREPARE_TEST - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4027	Station Keylock - Single Lock - PREPARE_TEST - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4023	Station Keylock - Single Lock - PREPARE_TEST - single input channel change - switch to SHORT_CIRCUIT_TEST	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4030	Station Keylock - Single Lock - PREPARE_TEST - multi input channel change - switch to SHORT_CIRCUIT_TEST	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4032	Station Keylock - Single Lock - PREPARE_TEST - single input channel change - ignore	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4024	Station Keylock - Single Lock - SHORT_CIRCUIT_TEST - Release / Cancel command - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4025	Station Keylock - Single Lock - SHORT_CIRCUIT_TEST - RELEASE_TIMER expiry - switch to PREPARE_RELEASE	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4026	Station Keylock - Single Lock - SHORT_CIRCUIT_TEST - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4028	Station Keylock - Single Lock - PREPARE_RELEASE - Release / Cancel command - do nothing	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
SEA-KLS:SSS:REQ: 4029	Station Keylock - Single Lock - PREPARE_RELEASE - RELEASE_TIMER expiry - switch to RELEASING	Functional Requirement	draft
SEA-KLS:SSS:REQ: 4031	Station Keylock - Single Lock - PREPARE_RELEASE - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4033	Station Keylock - Single Lock - RELEASING - Release / Cancel command - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4034	Station Keylock - Single Lock - RELEASING - RELEASE_TIMER expiry - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4035	Station Keylock - Single Lock - RELEASING - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4036	Station Keylock - Single Lock - RELEASING - input channel change - switch to RELEASED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4037	Station Keylock - Single Lock - RELEASED - Release command - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4038	Station Keylock - Single Lock - RELEASED - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4039	Station Keylock - Single Lock - RELEASED - Cancel command - switch to PREPARE_LOCKED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4040	Station Keylock - Single Lock - RELEASED - input channel change - switch to WITHDRAWN_START	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4041	Station Keylock - Single Lock - WITHDRAWN_START - Cancel command - reject	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4057	Station Keylock - Single Lock - WITHDRAWN_START - Release command - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4042	Station Keylock - Single Lock - WITHDRAWN_START - unexpected input channel changes - switch to WITHDRAWN_DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4043	Station Keylock - Single Lock - WITHDRAWN_START - WITHDRAWN_TIMER expiry - switch to WITHDRAWN_DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4044	Station Keylock - Single Lock - WITHDRAWN_START - input channel change - switch to WITHDRAWN	FUR - Functional Requirement	draft
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SEA-KLS:SSS:REQ: 4045	Station Keylock - Single Lock - WITHDRAWN_DISTURBED - Release / Cancel command - reject	Functional Requirement	draft
SEA-KLS:SSS:REQ: 4046	Station Keylock - Single Lock - WITHDRAWN_DISTURBED - input channel changes - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4047	Station Keylock - Single Lock - WITHDRAWN_DISTURBED - input channel changes - switch to WITHDRAWN	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4048	Station Keylock - Single Lock - WITHDRAWN - Cancel command - reject	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4058	Station Keylock - Single Lock - WITHDRAWN - Release command - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4049	Station Keylock - Single Lock - WITHDRAWN - unexpected input channel changes - switch to WITHDRAWN_DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4050	Station Keylock - Single Lock - WITHDRAWN - input channel change - switch to PREPARE_LOCKED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4051	Station Keylock - Single Lock - PREPARE_LOCKED - Cancel command - reject	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4052	Station Keylock - Single Lock - PREPARE_LOCKED - Release command - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4061	Station Keylock - Single Lock - PREPARE_LOCKED - input channel changes - switch to LOCKING	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4053	Station Keylock - Single Lock - PREPARE_LOCKED - OUT_CMD_DELAY_TIMER expiry - switch to LOCKING	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4054	Station Keylock - Single Lock - PREPARE_LOCKED - OUT_CMD_DELAY_TIMER expiry - do nothing	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4055	Station Keylock - Single Lock - PREPARE_LOCKED from RELEASED / WITHDRAWN - OUT_CMD_DELAY_TIMER - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4066	Station Keylock - Single Lock - PREPARE_LOCKED from RELEASED - OUT_CMD_DELAY_TIMER - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4059	Station Keylock - Single Lock - PREPARE_LOCKED from LOCKED - input channel change - do nothing	FUR - Functional Requirement	draft
	Station Keylock - Single Lock - PREPARE_LOCKED from	FUR -	

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SEA-KLS:SSS:REQ: 4060	LOCKED - OUT_CMD_DELAY_TIMER - unexpected input channel changes - switch to DISTURBED	Functional Requirement	draft
SEA-KLS:SSS:REQ: 4112	Station Keylock - Single Lock - Reject command	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4210	Station Keylock - Single Lock - state transfer	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4540	Line Keylock Client - Ignore commands e_key_locked_p / e_key_acknowledged_p during update_history phase	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4541	Line Keylock Client - Execute command e_key_acknowledged_p during update_history phase in LOCKED - transition to ACKNOWLEDGED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4542	Line Keylock Client - Execute command e_key_locked_p during update_history phase in ACKNOWLEDGED - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4550	Line Keylock Client - General Inquiry request from CC	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4551	Line Keylock Client - CC connection loss - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4552	Line Keylock Client - [Re-]Connecting to OC-IO in DISCONNECTED - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4553	Line Keylock Client - [Re-]Connecting to OC-IO in DISCONNECTED - transition to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4554	Line Keylock Client - [Re-]Connecting to OC-IO in DISCONNECTED - transition to WITHDRAWN	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4555	Line Keylock Client - [Re-]Connecting to OC-IO in DISCONNECTED - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4556	Line Keylock Client - Connection loss to OC-IO - DISTURBED - transition to DISCONNECTED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4557	Line Keylock Client - Connection loss to OC-IO - transition to DISCONNECTED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4558	Line Keylock Client - Ignore command e_key_locked_p / e_key_acknowledged_p in DISCONNECTED - do nothing	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
SEA-KLL:SSS:REQ: 4300	Line Keylock Server - Initial state	Functional Requirement	draft
SEA-KLL:SSS:REQ: 4301	Line Keylock Server - LOCKED - Pre-Rel command - switch to PRE_RELEASED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4302	Line Keylock Server - LOCKED - unexpected input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4303	Line Keylock Server - LOCKED - checkback channel changes - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4305	Line Keylock Server - DISTURBED - input channel changes - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4306	Line Keylock Server - DISTURBED - input channel changes - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4309	Line Keylock Server - PRE_RELEASED - SLVFE_CB / SLFE_CB checkback channel change - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4310	Line Keylock Server - PRE_RELEASED - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4342	Line Keylock Server - PRE_RELEASED - switch to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4321	Line Keylock Server - PRE_RELEASED - switch to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4312	Line Keylock Server - PRE_RELEASED - transition to QT_PRESSED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4313	Line Keylock Server - QT_PRESSED - QT_SYNC_TIMER expiry - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4314	Line Keylock Server - QT_PRESSED - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4316	Line Keylock Server - QT_PRESSED - input channel change - switch to ACKNOWLEDGING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4317	Line Keylock Server - ACKNOWLEDGING - QT_SYNC_TIMER expiry - transition to DISTURBED	FUR - Functional Requirement	draft
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SEA-KLL:SSS:REQ: 4318	Line Keylock Server - ACKNOWLEDGING - input channel change - do nothing	Functional Requirement	draft
SEA-KLL:SSS:REQ: 4319	Line Keylock Server - ACKNOWLEDGING - checkback channel change - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4320	Line Keylock Server - ACKNOWLEDGING - unexpected input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4322	Line Keylock Server - ACKNOWLEDGING - input channel changes - transition to ACKNOWLEDGED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4324	Line Keylock Server - ACKNOWLEDGING - multiple input channel change - transition to ACKNOWLEDGED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4323	Line Keylock Server - ACKNOWLEDGED - unexpected input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4325	Line Keylock Server - ACKNOWLEDGED - Release norm command - switch to RELEASED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4327	Line Keylock Server - RELEASED - SLFE_CB checkback channel change - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4328	Line Keylock Server - RELEASED - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4330	Line Keylock Server - RELEASED - input channel changes - transition to WITHDRAWN	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4331	Line Keylock Server - WITHDRAWN - WD_TIMER expiry - transition to LOCKING	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4332	Line Keylock Server - WITHDRAWN - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4334	Line Keylock Server - WITHDRAWN - input channel change - transition to WITHDRAW_RETRY	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4307	Line Keylock Server - WITHDRAW_RETRY - WD_RETRY_TIMER expiry - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4308	Line Keylock Server - WITHDRAW_RETRY - unexpected input channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
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SEA-KLL:SSS:REQ: 4315	Line Keylock Server - WITHDRAW_RETRY - input channel change - transition to WITHDRAWN	Functional Requirement	draft
SEA-KLL:SSS:REQ: 4335	Line Keylock Server - LOCKING - SLFE_CB checkback channel change - do nothing	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4336	Line Keylock Server - LOCKING - unexpected input channel changes - transition to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4338	Line Keylock Server - LOCKING - input channel changes - transition to LOCKED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4337	Line Keylock Server - Reject commands	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4304	Line Keylock Server - SLVFE_CB checkback channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4311	Line Keylock Server - SLFE_CB checkback channel changes - switch to DISTURBED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4343	Line Keylock Server - Reboot when e_key_locked_p command is received	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4560	Line Keylock Client - aspect query - OC-IO CONNECTED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4561	Line Keylock Client - aspect query - DISCONNECTED - OC-IO NOT_CONNECTED	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4460	Line Keylock Server - state transfer	FUR - Functional Requirement	draft
SEA-KLL:SSS:REQ: 4570	Line Keylock Client - state transfer	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4201	Station Keylock - Single Lock - aspect query - stable state - OC-IO CONNECTED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4203	Station Keylock - Single Lock - aspect query - LOCKING - last stable state DISCONNECTED / DISTURBED / LOCKED / RELEASED / WITHDRAWN - OC-IO CONNECTED	FUR - Functional Requirement	draft
SEA-KLS:SSS:REQ: 4204	Station Keylock - Single Lock - aspect query - PREPARE_LOCKED - last stable state LOCKED / RELEASED / WITHDRAWN - OC-IO CONNECTED	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
SEA-KLS:SSS:REQ: 4202	Station Keylock - Single Lock - aspect query in DISCONNECTED - OC-IO NOT_CONNECTED	Functional Requirement	draft

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